Evaluating the Impact of Augmenting the VAEP Model with 360-Degree Freeze-Frame Spatial Context

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Abstract

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1. Introduction

The adoption of analytics in association football has grown significantly in the last couple of decades (Goes et al., 2021; Olthof & Davis, 2025; Schaerlaeckens, 2025; Wakelam et al., 2022), enabling football clubs to leverage insights to inform tactical decisions, coaching and recruitment. Much of this innovation has involved the development of metrics used to determine the value provided by a particular action made by a player in terms of goal-scoring probabilities. One such metric is Valuing Actions by Estimating Probabilities (VAEP) (Decroos et al., 2019).

VAEP considers the probability of scoring or conceding within the next k events to assign a value between -2 and 2 to each action. VAEP, like other such metrics, typically uses action-centric features such as the action's start and end location and type (carry, pass, shot) in estimating scoring/conceding probabilties, but this approach does not explicitly model the wider spatial context (Sá-Freire, 2025). The spatial context - the position of teammates and opponents around the actor - of an action determines the options available to a player, showing viable passing options or space to carry into, as well as the proximity of opposition defenders applying pressure or blocking options.

The StatsBomb Open Data repository (StatsBomb, 2026a) provides events (e.g. carries, passes, shots) data for matches, along with *three-sixty* freeze-frames which indicate the positions of players within proximity of the acting player. Such data can be used to incorporate spatial context as a predictive feature. To illustrate this, **Figure 1** compares a pass event from a match with and without the surrounding spatial context. Without the spatial context, this pass appears to offer limited progression towards the opponent's goal. Traditional VAEP would consider how passes of this trajectory tend to result in scoring/conceding, but the addition of spatial context differentiates this particular pass from others of similar trajectory. **Figure 1** shows how the left hand side of the actor is blocked by multiple opponents (red) and this pass moves the ball into space on the right flank to open an attacking opportunity (in this case, a long-ranged shot)[1]. Spatial context could allow finer-grained differentiation, rather than aggregating all passes of this trajectory.

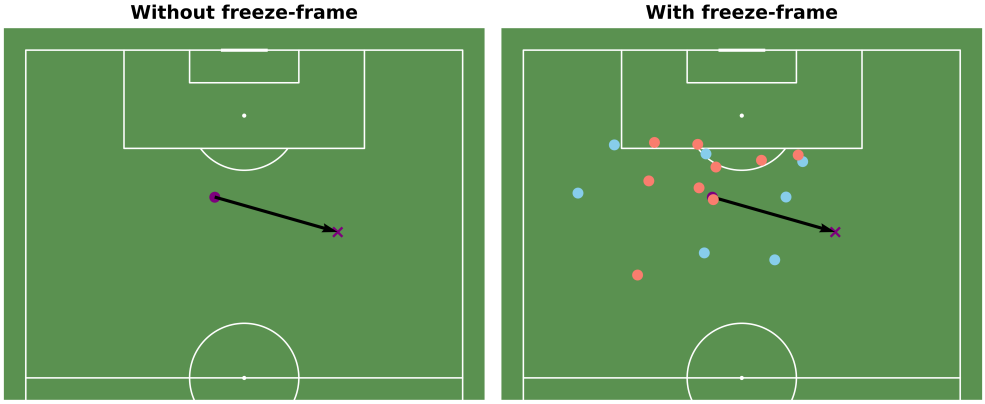


Figure 1: Pass Event With and Without Freeze-Frame

[1]The pass event id is '23b6b26a-293c-4d5e-9889-5a5ce0b7ad8c' and the subsequent long-ranged shot event id is 'd9d27d20-35bc-42bc-ac53-059ad519f781'

2. Research Objectives

The general aim of this project is to determine to what extent the addition of spatial context features enhances and alters the standard VAEP model. This entails the following objectives:

- Develop representations of spatial context from StatsBomb 360 freeze-frame data using feature engineering and deep learning techniques.
- Train and evaluate VAEP models, both with and without the inclusion of spatial context representations, using supervised classification algorithms.
- Compare and interpret these VAEP models using appropriate peformance metrics and an exploration of feature importance.

3. Background

In soccer analytics, *actions* refer to a *type of event* within a match which is caused by a player (e.g. a pass), with *events* being the broader set of significant occurrences in a match (e.g. passes, but also substitutions, fouls etc.) (Decroos, Bransen, Van Haaren, & Davis, 2019). Typical performance metrics focus on action outcomes, such as goals, assists and pass completion rates. Whilst such metrics can be used to determine the value of some actions and are useful for summarising player ability, soccer analytics has developed metrics to help generalise this notion of action value. Moreover, models used to assign value to actions have been enhanced by incorporating the locations of teammates and opponents, leading to improved performance and interpretability.

As such, this section will introduce the VAEP metric and outline how spatial context has been successfully incorporated into other action valuation metrics, motivating this project's research objectives.

3.1. Valuing Actions by Estimating Probabilities (VAEP)

Decroos, Bransen, Van Haaren, & Davis (2019) introduce Valuing Actions by Estimating Probabilities (VAEP). VAEP values actions in terms of the probability of scoring and the probability of conceding within the next k actions. Equation 1 defines the VAEP of an action a_i for team x.

$$V(a_i, x) = \Delta P_{scores}(a_i, x) - \Delta P_{concedes}(a_i, x), \quad V(a_i, x) \in [-2, 2] \tag{1}$$

If we consider a match as a sequence of m on-ball actions $[a_1, ..., a_m]$, we can view some game state S_i which corresponds to some subset of actions $[a_1, ..., a_i]$ and determine how the last action in the game state a_i changed the probability of team x scoring and conceding within the next k actions. The change in the probability of scoring (and conceding) from the previous game state $S_{i-1} = [a_1, ..., a_{i-1}]$ to the current game state S_i can be determined by simply finding their difference:

$$\begin{aligned} \Delta P_{\text{scores}}(a_i, x) &= P_{\text{scores}}(S_i, x) - P_{\text{scores}}(S_{i-1}, x) \\ \Delta P_{\text{concede}}(a_i, x) &= P_{\text{concede}}(S_i, x) - P_{\text{concede}}(S_{i-1}, x) \end{aligned} \tag{2}$$

These two probabilities are estimated using two supervised classification algorithms, one which estimates scoring, and the other conceding, within the next k actions. The feature sets used across these two models are identical, all that varies is the labels (scoring or conceding). Decroos, Bransen, Van Haaren, & Davis (2019) used a feature set which corresponds to the

Soccer Player Action Description Language (SPADL) format involving 9 features, most notably the action's start and end locations, and the type of action. The authors also add "complex features", such as the time elapsed between consecutive actions, as well as "game context features", such as the scoreline at the time of the action and the amount of time remaining.

Decroos, Bransen, Van Haaren, & Davis (2019) initially compared logistic regression, a random forest, XGBoost and CatBoost, finding CatBoost to perform best across both the scoring and conceding models in terms of the Brier Score and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Decroos, Bransen, Van Haaren, & Davis (2019) also extend the VAEP metric by using it to provide player ratings, where the VAEP of each action in a player's set of actions A_p^T across some time frame T are considered cumulatively, and normalised per-90 minutes played during T.

$$\text{rating } (p) = \frac{90}{m} \sum_{a_i \in A_p^T} V(a_i) \tag{3}$$

The authors compare these VAEP player ratings against traditional metrics like goals and assists per-90, as well as market valuations of players. The authors found the top-10 highest players ranked by VAEP captured players typically considered to be amongst the best in the Premier League, despite them not appearing in the top-10 goals + assists per-90 ranking. The top-10 VAEP ranked list also had a higher overall market valuation than the top-10 goals + assists per-90 ranked list (€$1,110$ Million vs $947 Million$), which suggests VAEP more comprehensively represents player value compared to traditional metrics.

Decroos & Davis (2020) extend their previous work by improving model interpretability. Given that the initial findings from Decroos, Bransen, Van Haaren, & Davis (2019) suggest the use of gradient boosting algorithms for VAEP model classifiers, these models are more difficult to interpret than methods such as logistic regression. As such, Decroos & Davis (2020) built a generalised additive model (GAM) on the top-10 most important features from the original VAEP feature set, as determined by the average *gain* (for the present classifiers, this is the reduction of Gini impurity - see Section 4.6.5) that feature imports on the model across all splits involving that feature. The authors found a very small drop in classifier performance: a 0.2% increase in the normalised Brier score compared to XGBoost on the original VAEP feature set, and a 0.1 increase on the top-10 most important feature set. The authors argue the large gains in utility came from the added interpretability from being able to assess each univariate function which comprised their GAM model. This enables a direct observation of how each function contributes to its prediction across the domain of that function. For example, *distance to opponent goal* constituted one of the top-10 features, and figure 2 in Decroos & Davis (2020) shows how the goal scoring probability changes across the domain of this feature's function. Matching intuition, a higher goal scoring probability is observed for lower distances to the opponent's goal, with this probability dropping exponentially as distance increases. This added interpretability helps to demonstrate the greater tactical understanding which soccer analysts may benefit from when using a GAM-based VAEP model, and that such interpretability may be gained without a significant loss in performance.

One potential limitation of the VAEP model is that it lacks a representation of wider spatial context from the positions of teammates and opponents during actions (Sá-Freire, 2025), due to its use of event data only. Every action taken by a player exists within a spatial context which determines the options available to a player, influencing their decision making. The use of spatial context in VAEP models is also understudied, despite its addition having been successfully applied to a number of alternative action value metrics.

3.2. Related Work: Applications of Spatial Context to Action Value Metrics

3.2.1. *Expected Goals (xG)*

Expected goals (xG) is an approach commonly used for assigning value to individual shots (Green, 2012; Macdonald, 2012). General to xG approaches is the representation of the value of some shot as the probability that it results in a goal. These goal probabilities are estimated via various classification algorithms such as logistic regression, gradient boosting and deep learning methods (Mead et al., 2023). Features used in xG models typically include *action-intrinsic* variables such as distance/angle to the centre of the opponent's goal and the body part used, along with *match-context* variables such as the current match scenario (e.g. set piece, open play) (Anzer & Bauer, 2021; Mead, O'Hare, & McMenemy, 2023; Robberechts & Davis, 2020).

Cefis & Carpita (2025) incorporate spatial context into the xG model by considering the number of players ahead of the shooter (x co-ordinate greater than the shooter's) within a fixed radius (y co-ordinate $\pm 10\%$ of the shooter's). From this count, they derive two binary variables `D1.OpponentsPlayer` and `D2.OpponentsPlayer`, where the former equals 1 when the number of players in this area is greater than 0, and the latter equals 1 when the number of players in the area is greater than 1. They also consider the x and y co-ordinates of the opponent's goalkeeper (`GK x coordinate`, `GK y coordinate`). Table 3 in Cefis & Carpita (2025) show the coefficients and odds ratios of each feature in their logistic regression xG model across their 4 different techniques for addressing class imbalance. `D1.OpponentsPlayer`, `D2.OpponentsPlayer`, and `GK x coordinate` all have negative coefficients, indicating a reduction in the log odds of the shot resulting in a goal. This is particularly pronounced for `D2.OpponentsPlayer`, with a coefficient range of -6.13 to -5.69, indicating that the probability of scoring (i.e. xG) is reduced by around 99.73% (because $1 - e^{(-5.69 + -6.13)/2} \approx 0.9973$) when there is more than 1 opposition defender within proximity of the shooter. This finding matches intuitions about the influence of the opposition's positioning on goal probability and highlights how the addition of spatial context during modelling can enhance the xG metric.

3.2.2. *Expected Threat (xT)*

A further action value metric is expected threat (xT). xT measures the value of actions based on where on the pitch the action moves the ball from and to. Singh (2018) derived the following equation for calculating the xT of a given location on the pitch:

$$\mathrm{xT}_{x,y} = \left(s_{x,y} \times g_{x,y}\right)\left(m_{x,y} \times \sum_{z=1}^{Z}\sum_{w=1}^{W} T_{(x,y)\to(z,w)}\mathrm{xT}_{z,w}\right) \tag{4}$$

If we discretise the football pitch into some $Z \times W$ (Singh (2018) uses $Z = 16$ and $W = 12$) set of equally-sized square zones (x, y), we can consider three possible attacking options when the ball is in some zone (x, y): shoot, carry or pass. $\left(s_{x,y} \times g_{x,y}\right)$ quantifies the value of shooting from zone (x, y) by weighting the probability of scoring $g_{x,y}$ by the probability of shooting $s_{x,y}$. The other two options, carrying or passing, involve moving the ball from (x, y) to (z, w). The transition matrix $T_{(x,y)}$ provides the probablity of moving the ball from zone (x, y) to any zone (z, w) after 1 time step. If we consider all $Z \times W$ possible options and multiply their transition probabilities by the value of its respective (z, w) zone, we can take their sum and weight it by the probability of actually moving the ball to another zone when in (x, y). This provides a quantification of passing or carrying the ball, which when multiplied by the shot-probability weighted xG of zone (x, y) provides the xT of the ball being at zone (x, y). However, given that Equation 4 assumes that the value of each zone $\left(\mathrm{xT}_{z,w}\right)$ is already known a-priori, we must

estimate its value. The xT model thus begins by setting every zone (x, y) to have an xT of 0. This simplies Equation 4 to essentialy be the xG of zone (x, y), weighted by how likely players are to actually shoot from that zone:

$$\mathrm{xT}^{(0)}_{x,y} = \left(s_{x,y} \times g_{x,y}\right) \tag{5}$$

We can then take the xT values from this first iteration and recalculate them iteratively until convergence. Singh (2018) suggests 4–5 iterations are sufficient but that this may vary across datasets. To then assign xT to an action, we simply subtract the xT of the end location of the action (i.e. the xT of zone (z, w)) by the xT of the start location (the xT of zone (x, y)). This approach is useful for allowing us to determine the value of any on-ball attacking action, not just shots. We can also quantify genuinely valuable actions at any point during a chain of actions (e.g. an incisive pass during a sequence which ends in a goal), rather than suffering from a bias towards teminating actions (goals, assists).

Everett et al. (2022) introduce spatial context to the xT transition matrix $T_{(x,y)}$ via a pitch control (Spearman et al., 2017) and interception model (Burriel & Buldú, 2021) which was used to train a convolutional neural network (CNN) to produce the transition probabilities for their contextual transition matrix $T^{\star}_{(x,y)}$. The pitch control model considers the probability of a player controlling the ball as a function of their distance from opposition players, whilst the interception model introduces an additional transition probability, that being the probability of the ball being intercepted when moving from zone (x, y) to its intended destination (z, w). Everett, Beal, Matthews, Norman, & Ramchurn (2022) found the introduction of spatial context yielded a modest improvement in terms of model fit, reducing the log likelihood of the xT model from -5.137 to -4.151. Greater gains were found in its useful applications to tactical analysis, such as enabling the authors to apply a Basin-Hopping optimiser (Wales & Doye, 1997) to find more optimal defensive shapes than which actually occurred during actions, using their derived Threat Above Expected (TAx) metric as the algorithm's loss function.

3.2.3. *Expected Possession Value (EPV)*

Expected possession value (EPV) (Fernandez et al., 2021) is an action valuation metric, derived from other sports such as basketball (Cervone et al., 2016), which exploits tracking data to assign value to a game state (e.g. during an action) at time t via a consideration of all possible subsequent paths it can take.

$$\mathrm{EPV}_t = \sum_{a \in A} \mathbb{E}[G \mid A = a, T_t] \quad \mathbb{P}(A = a \mid T_t) \tag{6}$$

Equation 6 shows that the EPV at time t for some action is determined by the expected value of G (typically a goal being scored or not) given some action a has occurred in some spatiotemporal context T_t, multiplied by the probability of that action a occurring in the spatiotemporal context T_t. For each action a considered in the action set A, we sum these expected values to provide an EPV of some game state at time t. Thus EPV can be applied to any game state, not simply at the time at which discrete match events or actions occur.

We can see from the term T_t that EPV relies upon spatial context to operate. Fernandez, Bornn, & Cervone (2021) list a number of spatial features used in the EPV model, such as pitch control, number of players within 3m of the ball location, and the number of players within the area circumscribed by the ball and the two posts of the opponent's goal. EPV is thus inherently sensitive to spatial context in its valuation of actions. The utility provided by EPV's

incorporation of spatial context is explored by Fernandez, Bornn, & Cervone (2021). For example, they use EPV to determine the on-ball (passes with positive EPV) and off-ball (pitch locations with positive EPV if a pass were to arrive there) advantage Liverpool in 2014/15 had during possession when up against different opponent formations. This analysis shows both where Liverpool were successful and unsuccessful in their passing decision (on-ball) and also where valuable regions of the pitch are to direct passes towards during build-up (off-ball). Such an analysis is extremely useful for tactical decision making as, in response to opposition pressing structures, teams can evaluate where on the pitch they should look to progress the ball towards to be most effective during build-up play, and to evaluate how successful their current approach is.

3.2.4. *Summary*

To summarise, the discussed applications demonstrate the utility provided by incorporating spatial context into action valuation models. Given the lack of investigation into augmenting VAEP models with wider spatial context, this project addresses this research gap and explores what role representations of spatial context play in altering model behaviour and, potentially, performance.

3.3. Representing Spatial Context

3.3.1. *Manual Spatial Feature Engineering*

Previous research has represented spatial context using features engineered from spatial data.

As described in Section 3.2.1, Cefis & Carpita (2025) derived two binary features `D1.OpponentsPlayer` and `D2.OpponentsPlayer` from player x and y co-coordinates, indicating the number of players within the immediate frontal proximity of the shooter.

Peters et al. (2024) developed an Expected Pass Turnovers (xPT) metric, used to measure the probability that a given pass would result in a turnover (conceding possession to the opposition) which can be used to assess how player's passing ability during some time interval compares to what would be expected on average, given the set of passes made by that player and the contextual factors relating to those passes. The authors derive xPT values from a binary logistic mixed-effects model trained on a feature set which included two features derived from StatsBomb *three-sixty* freeze-frames: "three pressure radii" (the number of opponents within three radii of varying, but unspecified length around the ball carrier) and number of "unmarked teammates" (teammates with no opponents within 2m of them). The authors found the addition of these "positional" features improved model AUC by 6.1 points

Amichay et al. (2025) measured the areas of convex hulls and inner layers to represent team shapes. A convex hull, in this context, is the area inscribed between the outer-most positioned players on the pitch for a given team. A larger area corresponds to a wider/longer team shape, whereas a smaller area is narrower/shorter. Since this convex hull only considers the outer-most players, a second convex layer can be created from the players who do not contribute to the hull. The ratio of these two areas forms the author's derived Layer Ratio (LR) metric, which summarises the compactness of the team shape.

3.3.2. *Spatial Embeddings: Set Transformer*

An alternative way to represent spatial context is in the form of spatial embeddings. A spatial embedding is a real-valued vector used to represent a set of geometric objects (in this case, player co-ordinates and their characteristics) in a fixed d-dimensional space. These embeddings

can represent non-linear relationships between objects and thus may serve to represent patterns, like defensive and offensive team structures during actions.

However, as seen in Figure 2, StatsBomb *three-sixty* data does not identify individual players within each freeze frame (except from the actor). Thus, to create embeddings for player positions, we must use a neural network which has an input layer which is *permutation invariant*. That is, the output of the network (i.e. the embeddings) should not depend on the order in which features appear in the input layer. This would enable the locations of players in each freeze-frame to be inputted into the network in any order and still produce the same embedding. It would also be useful for such a network to attend to spatial relationships that may exist across freeze-frames, such as defensive shapes.

One useful technique for developing a permutation-invariant network for spatial embeddings is a *set transformer* (Lee et al., 2019). A set transformer is a variation on the standard transformer architecture from Vaswani et al. (2017) which can be used to process sets such that the output of the transformer is invariant to how elements in the set are permuted when entering into the model.

Transformers use multihead self-attention layers in a traditional encoder-decoder architecture. Multihead self-attention works by projecting an input sequence $X \in \mathbf{R}^{n \times d}$ into different learned linear projections of the input embeddings: $\left\{Q = XW_Q, K = XW_K, V = XW_V\right\} \in \mathbb{R}^{n \times d}$. Each row corresponds to a token representation in a different learned subspace. The self-attention mechanism determines how much each token should attend to other tokens, including itself, by computing similarity between so-called *query* vectors and *key* vectors. If the similarity between a *query* and a *key* is high, the corresponding *value* vector contributes more strongly to the output representation. This is computed for all tokens simultaneously, producing pairwise interactions between all elements in the sequence, including self-interactions. The attention mechanism achieves this via scaled dot-product attention:

$$\text{Attention } (Q, K, V) = \text{softmax } \left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{7}$$

Multihead attention then involves performing this scaled dot-product attention in parallel h times, each with their own set of Q, K, V matrices. Outputs are then concatenated and fed into a standard feedforward neural network.

The multi-head attention mechanism of standard transformers is adapted in set transformers via *induced set attention blocks* (ISABs). An ISAB performs self-attention between set elements by introducing *inducing points* which contain parameters learned during training. These inducing points are some fixed-sized vector I and attend to the inputted set X to produce a transformed set H. The input set X then attends to H to produce a final set of n elements. This method improves computational efficiency as the size of the inducing points vector $|I|$ can be set to be smaller than the input size $|X|$, thus avoiding standard pairwise self-attention and reducing time complexity from $O(n^2)$ to $O(nm)$ where $m < n$.

Important for the purposes of this present project, set transformers also enable permutation invariance by dropping position encodings of input set elements and pool outputs via pooling with multihead attention (PMA). PMA works similarly to multihead attention but instead pools the representations of each element of the set into k vectors, each providing a representation of the whole set. k is typically set to 1, provding a single representation of the set, but $k > 1$ is permitted. The queries Q are a fixed set of learned *seed vectors* S. Reordering the set will thus result in the same pooled representation via PMA, enabling permutation invariance.

As such, the encoder component of the set transformer is used to generate spatial embeddings. However, regarding the decoder, this component is simply used during the training process. The decoder takes the encoder's generated spatial embedding as an input and attempts to reconstruct the original freeze-frame array. The deviation between the original array and the decoder's array provides a means to assess how well the encoder's generated spatial embedding represents the freeze-frame array, and thereby constitutes a measure of the set transformer's performance: its loss function. The exact loss function used in this present project is discussed in Section 4.4.2.1. Given that the role of the decoder is to provide a reconstruction objective for evaluating the quality of the encoder's learned spatial embedding, it does not necessarily require the same self-attention mechanisms used by the encoder. Such a design choice to use a lightweight, non-attention based decoder has been applied in the SegFormer architecture (Xie et al., 2021) for regular transformers, and successfully applied to set transformers by Ashcroft et al. (2023) in their work on generating illustrations by training a generative set transformer model to represent them as "stroke clouds" (a type of set comprised of Bézier curves) whilst using a non-attention based diffusion decoder. In this present project, the encoder is responsible for modelling relationships between players and producing a fixed-dimensional representation of the freeze frame, while the decoder acts as a reconstruction head that maps this representation back to the original player freeze-frame. With the decoder playing an instrumental role in the development of spatial embeddings, it is arguably sufficient for the decoder to have a relatively simplistic architecture, provided the loss function used preserves permutation-invariance.

Applications of set transformers in soccer analytics have typically fallen in the domain of trajectory prediction. Kim et al. (2023) apply set transformers in the context of soccer analytics. Kim, Choi, Kim, Yoon, & Ko (2023) use set transformers to encode representations of player trajectories from tracking data snapshots to infer ball trajectories. Capellera et al. (2024) similarly developed *FootBots* using set transformer architecture to improve player and ball motion prediction over traditional models. Whilst this domain deviates from this project, these studies do highlight the appropriateness of using set transformers to encode player positions.

This project seeks to use set transformers to produced fixed-sizes spatial embeddings for each team's set of players within each freeze-frame sample. This can be achieved by first converting each individual co-ordinate pair, along with its characteristics (teammate, goalkeeper, actor), into an embedding via an initial input projection sublayer. Each set of individual co-ordinate embeddings for each team in each freeze-frame can then be passed as an input tensor to the set transformer which will produce embedding representations of each set of co-ordinates. The initial embedding of co-ordinate pairs is necessary to ensure the set transformer has enough dimensions to learn from whereas simply having 5 dimensions (i.e. x, y co-ordinates, and the 3 player attributes) would be insufficient to capture multiple non-linear relationships.

4. Methodology

4.1. Datasets Outline

The project uses data from StatsBomb's Open Data repository. This repository has "made certain leagues of StatsBomb Data freely available for public use for research projects and genuine interest in football analytics" (StatsBomb, 2026a). The StatsBomb Open Data repository provides the following datasets, and data specification documents exist which outline variables, data types and variable descriptions for each dataset:

- **competitions.json:** A single JSON file for competition summary information.
- **events:** A directory containing a single JSON file for each match, detailing each event which takes place within that match (e.g. each pass, shot, carry etc.).

- **lineups:** A directory containing a single JSON file for each match, providing information about each player who participated in that match (e.g. their position, shirt number, nationality).
- **matches:** A directory containing a single JSON file for a season in a given competition, providing summary information about each match played in that season (e.g. match date, home/away teams and scores).
- **three-sixty:** A directory containing a single JSON file for each match, providing data for each event in a given match detailing the position of each player in the broadcaster's frame at the moment of that event.

Decroos, Bransen, Van Haaren, & Davis (2019), along with VAEP, introduce the Soccer Player Action Description Language (SPADL) as a data engineering format to enable "human-interpretable, simple and complete" records of actions within football matches. A variant of this format is used in this project for the baseline feature set, that being Atomic-SPADL (Davis et al., 2020). Atomic-SPADL represents an action without considering its result (success vs failure), unlike SPADL which includes the result. An example of the effect this has on action representation is that a pass (one action under SPADL) is separated into two actions: the giving of the pass, and the reception of the pass. An additional "interception" action type is introduced to enable incomplete passes to be captured. Atomic-SPADL therefore introduces more precision in how actions are represented as actions are now indivisible (hence atomic). This atomic-SPADL format is then augmented in this project by the addition of engineered spatial features.

The project uses *events* and *three-sixty* data corresponding to 292 matches from UEFA Men's Euro 2020 & 2024, UEFA Women's Euro 2022 & 2025, FIFA Men's World Cup 2022, FIFA Women's World Cup 2023. Unlike most competitions in this repository, *three-sixty* data is available for every match in this sample, enabling player positional context to be incorporated into feature engineering for 1,052,116 match events.

4.1.1. *Limitations of Datasets*

Football analysts often makes use of tracking data (Goes, Meerhoff, Bueno, Rodrigues, Moura, Brink, Elferink-Gemser, Knobbe, Cunha, Torres, & Lemmink, 2021). Rather than considering discrete time events, tracking data provides a continuous record of player and ball position for the duration of a match. This enables analysts to model actions in matches as continuous processes, accounting for the position and movements of the entire team, unlike with events data. Such tracking data is often proprietory and is thus not available for public use (Bassek et al., 2025). This limits the scope of the present project to viewing events as discrete in time, thus losing some surrounding context which may be informative (e.g. player movement direction and velocity).

Events data is nonetheless relatively comprehensive, covering a much larger sample of matches than the limited publicly-available tracking data (Olthof & Davis, 2025). Furthermore, the use of StatsBomb's *three-sixty* data does address this limitation somewhat by providing snapshots of proximal player positions at the point of each event, allowing some inference to be made regarding team positioning during match events. However, these snapshots do not necessarily capture every player's location during an event, just those within the view of the broadcast camera (StatsBomb, 2026a).

A further limitation is with the use of SPADL formatted data. Recent work by Jo et al. (2026) has shown that some events data contains logical inconsistencies (i.e. sequences of events which are not logically possible), which the SPADL format does not correct for. The authors develop the Verified Event Data Format for Reliable Soccer Analytics (VERSA) to address this issue

using a state-transition model to automatically identify and correct logical inconsitencies, improving performance on VAEP model training. However, their results on StatsBomb data suggest that the atomic-SPADL format still slightly outperforms VERSA. This result, along with the increased precision in action representation mentioned earlier, motivates the use of the atomic-SPADL format in this project, but it is recognised that the potential data integrity issues highlighted by Jo, Kang, Lee, Lee, Bauer, & Ko (2026) will not be explicitly handeled as implementing VERSA is non-trivial and thus beyond the scope of this project.

4.2. Data Collection

Data was obtained from the StatsBomb Open Data Repository via the `statsbombpy` (StatsBomb, 2026b), `socceraction` (Socceraction Developers, n.d., Decroos, Bransen, Van Haaren, & Davis (2019)), and requests (, n.d.) libraries. The `statsbombpy` and `socceractions` libraries were used to collect raw events data from the matches in each competition, whilst `requests` was used to collect *three-sixty* freeze-frames from the StatsBomb Open Data Repository directly. Events data was then merged with freeze-frame data and each competition's data was saved to an individual parquet file.

4.3. Data Preprocessing

With StatsBomb events and *three-sixty* data collected, the first preprocessing step requires converting these events into SPADL-formatted actions. This was achieved by passing each match's set of events to the `convert_to_actions` function from `socceraction`. This also required the rejoining of freeze frame data, since this preprocessing function only returns SPADL formatted data, not custom features.

To then convert these actions to the atomic-SPADL format, the `convert_to_atomic` function similarly takes each match's set of actions and returns them in atomic-SPADL format. The atomic actions of each match from each competition are then saved to parquet files, one for each competition.

With the derived atomic-SPADL actions, spatial context features are subsequently added during data engineering.

4.4. Data Engineering

4.4.1. *Manually-Engineered Spatial Context Features*

```
{
    "teammate": boolean,
    "actor": boolean,
    "keeper": boolean,
    "location": [float, float]
}
```

Figure 2: StatsBomb *three-sixty* freeze-frame JSON schema

Figure 2 outlines the JSON schema for StatsBomb *three-sixty* freeze-frames. Each freeze-frame JSON object has an associated `event_uuid` to enable joins with events data, and a `visible_area` polygon which indicates the viewing area from the broadcast camera at the moment the action took place. We can see from the available data, that it is possible to construct the opposition

proximity metric from Cefis & Carpita (2025) and a partial convex hull area or LR as explored by Amichay, Silva, Brito, & Marcelino (2025). Given that the sample in this project applies to on-ball actions, not just shots, the opposition proximity metric from Cefis & Carpita (2025) is adapted to consider a 360° radius around the player, rather than a frontal semi-circle.

Thus, for this project, the following spatial features were engineered for teammates and opponents, respectively, using *three-sixty* data. These are both inspired by the literature and also chosen by myself:

- Euclidean distance to nearest teammate (meters)
- Euclidean distance to nearest opponent (meters)
- Number of teammates within 5 meters of the actor
- Number of opponents within 5 meters of the actor
- Average Euclidean distance to teammates in freeze-frame (meters)
- Standard deviation of Euclidean distance to teammates in freeze-frame (meters)
- Average Euclidean distance to opponents in freeze-frame (meters)
- Standard deviation of Euclidean distance to opponents in freeze-frame (meters)
- Teammate convex hull area (based on players within freeze-frame) (m^2)
- Opponent convex hull area (based on players within freeze-frame) (m^2)

To outline the rationale here, opponent spatial metrics are intended to provide a proxy measure of *pressure* applied to the actor and *defensive shape*, which imposes limitations upon where the ball can be progressed to by introducing a risk of losing possession. Teammate spatial metrics provide a proxy measure of *pass availability* for the actor and *rest defense* - the positional structure of a team whilst in possession to prepare for potential counter-attacks (Coaches' Voice, 2024).

For *number of teammates/opponents within 5 meters of the actor*, whilst a radius of 5 meters somewhat arbitrary, Bischofberger et al. (2026) argue that a "lower radius might risk missing actual pressure and involvement while a higher radius might lead to a weaker attribution between players and stronger intra-team correlations" p.5. Calabuig et al. (2024) also use a 5 meter radius around players as part of their construction of *influence areas* - the amount of overlap of two player's circles equates to how much influence the defender has over the attacker, providing a measure of defensive pressure.

It should be noted that such an approach to approximating opposition pressure or teammate pass availability is considered relatively simplistic, with more sophisticated approaches such as pitch control (Fernández et al., 2018) being preferred in the literature. However, given that metrics like pitch control require tracking data to be implemented correctly, they fall beyond this project's scope. Regardless, the intention with these manually-engineered spatial context features is to provide an approximation of spatial context, which is easy to implement, in order to contrast it with the generated spatial context embeddings in terms of classifier performance and behaviour. Given that such a measure has been successfully and usefully incorporated in previous literature, it was deemed appropriate for this present project.

4.4.2. *Spatial Context Embeddings*

To generate an n-dimensional embedding of each freeze-frame for each action, we can begin by converting each JSON object in each freeze-frame to a 1×5 array, using the 5 available features as shown in Figure 2 (the location key can be split into an x co-ordinate feature and a y co-ordinate feature, respectively). Each freeze-frame consists of n players, and so n arrays are stacked into an $n \times 5$ array.

This $n \times 5$ array, constituting each player within a single freeze-frame, is then be projected into a higher dimensional space via an initial learned affine transformation h_i.

$$h_i = W_e x_i + b_e, \quad x_i \in \mathbb{R}^5, h_i \in \mathbb{R}^{d_{\text{hidden}}} \tag{8}$$

The dimensionality d_{hidden} of this learned projection layer is left as a tunable hyperparameter during optimisation, with $d_{\text{hidden}} \in \{32, 64, 128, 256\}$ being the candidates used. This transformation provides the set transformer with a higher-dimensional latent representation of each element, enabling it to learn richer relationships between players within each freeze-frame than would be possible from the raw feature representation alone.

This projection layer $H \in \mathbb{R}^{n \times d_{\text{hidden}}}$ is then inputted to the set transformer encoder, whereby the self-attention mechanism learns the weight matrices W_Q, W_K, W_V which minimise loss (see Section 4.4.2.1).

4.4.2.1. *Set Transformer Loss Function*

Given that the freeze-frames - particularly the player co-ordinates - constitute permutation-invariant sets, the loss function used must also respect this characteristic. Each player in each freeze-frame consists of two broad components: a pair of x, y co-ordinates for their location on the pitch and a set of attributes about that player (teammate, goalkeeper, actor).

For the location component, we require a loss function which preserves the permutation-invariant property of our freeze-frames, which are unordered sets. Chamfer distance fulfills such requirements.

Chamfer distance conducts a nearest neighbour search between points in one set with points in the other set. Each point is assigned its nearest neighbour in the other set and the distance is taken between them. Each of these distances are summed and averaged across the number of elements from the first set. Minimisation of Chamfer distance is achieved by successfully reconstructing the original set. In this scenario, this would correspond to the decoder successfully reconstructing the co-ordinate locations of each player within the original freeze-frame. This nearest neighbour search can be conducted with quadratic time complexity, as we must compute every pairwise distance between original and reconstructed set elements.

$$\mathcal{L}_{\text{Chamfer}} = d_{\text{CH}}\big(S, \hat{S}\big) = \frac{1}{|S|} \sum_{x \in S} \min_{\hat{x} \in \hat{S}} \| \, x - \hat{x} \, \|_2^2 + \frac{1}{\big|\hat{S}\big|} \sum_{\hat{x} \in \hat{S}} \min_{x \in S} \| \, \hat{x} - x \, \|_2^2 \tag{9}$$

Alternative distance measures may be used in place of Chamfer distance, such as the Hungarian method (Kuhn, 1955), which can be used to find an optimal solution to the assignment problem: finding the permutation of the reconstructed set which, when paired one-to-one with the original set, minimises loss. Chamfer loss does not enforce such a one-to-one correspondence, as elements from the original set may share the same nearest neighbour from the reconstructed set (and vice versa).

Rather than searching through each permutation naively, which would result in a factorial search time complexity, the Hungarian method constructs a cost matrix which records the pairwise distances of each point in the original set with each point in the reconstructed set.

To find the optimal permutation, the minimum value of each row is subtracted from its elements. Then, the minimum value of each column is subtracted from each of its elements. If each row in the resulting matrix has an element equal to 0 where the column (j-index) of that

element is not reused in the 0-valued elements of any other row, then an optimal assignment can be derived.

If this is not the case, we find the number of horizontal and vertical lines needed to cover each 0 element (a *covered value*) in the matrix. If less than n lines were used, where n is the number of rows and columns in the cost matrix, then we subtract the minimum positive-valued row value from each row element which is *uncovered* (i.e. non-zero) and add the value to elements which are covered. We then reassess whether n lines are needed to cover the 0 values and repeat until this condition is met. The resulting matrix elements with values equal to 0 correspond to the optimal one-to-one pairing of elements in the original set with elements in the reconstructed set. This approach to the Hungarian method has a cubic time complexity, resulting in a substantial improvement over the naive method.

Despite the Hungarian method having the advantage of preserving a one-to-one correspondence between original and reconstructed set elements, Chamfer loss was deemed preferable for this project as:

- Reconstructed freeze-frames are used as a training signal to produce encoder-generated embeddings which capture meaningful spatial context information. The decoder-generated freeze-frames are not needed for any downstream tasks (e.g. synthetically generating realistic football player event freeze-frames).
- Players in freeze-frames are not uniquely identifiable, so enforcing one-to-one correspondence does not necessarily provide additional semantic information.
- Chamfer loss operates with $O(n^2)$, whereas Hungarian matching is $O(n^3)$. For a training set size $n \in [10^5, 10^6]$, such as in this project, and given that the constant factor of Chamfer distance is comparatively computationally cheaper than that of the Hungarian method, the reduction of time complexity by a factor of n makes using Chamfer loss more economical.

Future work with developing spatial embeddings from freeze-frames may wish to empirically contrast alternative reconstruction loss functions whilst considering the trade-off in terms of computation time.

The second component of this set transformer's loss function relates to the teammate, actor and goalkeeper flags which each player in each freeze-frame as, as seen in Figure 2. Given that the teammate, actor and goalkeeper statuses of each player are binary categorical variables, we can model their realisations as being sampled from Bernoulli distributions. The decoder can therefore predict a probability of the flag being equal to 1 and we can use this prediction to evaluate performance.

$$\mathcal{L}_{\text{BCE}} = \frac{1}{N}\sum_{i=1}^{N}(-y_i \log \hat{p}_i - (1-y_i)\log(1-\hat{p}_i)) \tag{10}$$

Binary cross entropy (BCE) loss provides a measure of how close predicted probabilties are to the true binary values. When the true value is 1, the second term evaluates to 0 and thus the loss is determined by the first term $-y_i \log \hat{p}_i$, which evaluates to 0 when the predicted probability is 1, and approaches ∞ when the limit of the predicted probability tends to 0. When the true value is 0, the first term cancels and the second term similarly provides a measure of loss which is 0 when the predicted probability is 0, and approaches ∞ when the loss approaches 1. This loss is calculated for each data instance individually and its average is taken to provide an overall measure of BCE loss.

To combine these two measures into a single loss function for the set transformer, we can use a weighted sum, where the weights λ_{xy} and λ_f are tunable hyperparameters. Equation 11 therefore constitutes the loss function used for this present project's set transformer.

$$\mathcal{L} = \lambda_{xy}\mathcal{L}_{\text{Chamfer}} + \lambda_f\mathcal{L}_{\text{BCE}} \tag{11}$$

4.5. Set Transformer Hyperparameter Tuning

Rather than training a single set transformer with arbitrary hyperparameters, it would be preferable to use an algorithm to choose values for the model's hyperparameters as to minimise its loss function, whilst avoiding overfitting. This process is known as hyperparameter optimisation or tuning.

One standard approach to hyperparameter tuning is grid search. With grid search, a discrete set of candidate values are provided for each model hyperparameter and each possible combination of these candidates is trialled. Grid search, whilst exhaustive of the search space, is an inefficient means of hyperparameter tuning, relative to alternative methods. With h hyperparameters in the model, and v candidate values across all hyperparameters, the search space with grid search is equivalent to v^h, which constitutes an exponential time complexity $O(v^h)$. Grid search's efficacy is further compounded by the fact that much time is spent exploring hyperparameter value combinations which yield poor performance, relative to the optimal combination available. Moreover, the search space provided must be discrete, which potentially misses more optimal hyperparameter values that may be found in a continuous search space. Whilst potentially useful for small models where there exists well-established prior expectations for appropriate hyperparameter values, alternative methods may be preferred.

An alternative approach can be found in random search. Random search draws n random samples of hyperparameter value combinations from the set of values provided and trials them. Rather than an exhaustive search of a discrete set of hyperparameter value combinations, random search enables sampling over both discrete sets of values and also continuous sampling distributions for hyperparameter values. Compared to grid search, this helps to reduce the time complexity of the search space to linear time, with search time simply being a linear function of the number of desired trials - $O(n)$ - whilst also allowing for more optimal values to be chosen via the use of hyperparameter value sampling distributions. Bergstra & Bengio (2012) argue that since hyperparameters for a given model often do not contribute evenly to the performance of the model (e.g. the learning rate may be more important that the dropout rate), grid search wastes time exploring hyperparameter value combinations which will not yield significant gains in performance. Instead, random search randomly samples values for all hyperparameters at each trial, meaning that the most important hyperparameters are always being tested with new unique values within a fixed computational budget. Grid search instead wastes computational budget on varying less important hyperparameters whilst keeping important hyperparameters constant, since it must exhaust all possible combinations. Random search therefore can explore the search space of important hyperparameters more efficiently, providing opportunity for random search to outperform grid search.

Despite this advantage over grid search, a limitation common to both grid search and random search is the fact that neither approach uses information gained from previous trials to improve the estimations made by future trials. Sampling efficiency could be drastically improved if an algorithm were to consider previous trials in its sampling of new candidate values.

One family of optimiser approaches which we can use to enhance hyperparameter sampling efficiency are Bayesian optimisers. In general, a Bayesian optimiser constructs a surrogate model to predict model performance y given a set of hyperparameter values x. This surrogate model

$p(y \mid x)$ produces its estimates based on some prior history of trials. After each trial, the predictions from this surrogate model are used to inform an acquisition function, which explores the search space to find regions which balance currently acquired knowledge on which hyperparameters lead to improved performance over less explored regions where uncertainty is higher and thus potential for further optimisation may exist. This acquisition function then samples hyperparameter values based on this tradeoff, which in turn updates the surrogate model and thus the process continues for some n number of trials. A Bayesian optimisier can therefore incorporate prior evidence acquired from previous trials to inform decisions on which hyperparameter values ought to be trialled next, enabling convergence towards a local optimium.

The present project will use a type of Bayesian optimiser known as a Tree-Structured Parzen Estimator (TPE), which is outlined in the next subsection (Section 4.5.1). This TPE is implemented via the Optuna Python library (Akiba et al., 2019). Table 1 provides a brief outline of the hyperparameters to be used for the set transformer in this project.

Hyperparameter	Description
Batch Size	The number of data instances processed before model parameters are updated.
Dropout	A random proportion of network activations set to zero during training.
Embedding Dimensions	The number of dimensions in the output layer of the set transformer encoder.
Flag Weight	A weighting applied to the Chamfer distance component of the set transformer loss function.
Hidden Dimensions	The number of dimensions of each hidden layer.
Learning Rate	A weighting applied to the derivative of the loss function with respect to model parameters to control the magnitude by which parameters are updated during training.
Number of Heads	The number of self-attention mechanisms which operate in parallel in each hidden layer.
Number of Hidden Layers	The number of hidden layers used within the set transformer encoder.
Weight Decay	An L2 regularisation applied to model weights.
XY Weight	A weighting applied to the binary cross-entropy component the set transformer loss function.

Table 1: Set Transformer Hyperparameter Descriptions

4.5.1. *Tree-Structured Parzen Estimatior (TPE)*

A TPE is a type of Bayesian optimiser which deviates from the general approach in a few key ways. Firstly, a TPE models the probability distribution of hyperparameter values, given model performance $p(x \mid y)$ - an inversion of the standard Bayesian optimiser approach. Thus, a TPE seeks to update the probability distributions of each hyperparameter in line with the prior evidence accumulated about which values lead to improved performance. $p(x \mid y)$ is defined by two derived densities: $l(x)$ and $g(x)$. Given a set of trials with candidate hyperparameters x and

observed loss values y, a parameter γ is defined so as to dervied the γth quantile of y referred to as y^* such that $p(y < y^*) = \gamma$. The exact value of γ may be a fixed value (i.e. always use $\gamma = 0.15$), or as in the case with Optuna's default implementation, an adaptive gamma function:

$$\gamma(n) = \min(\lceil 0.1n \rceil, 25) \tag{12}$$

This threshold y^* is used to define two probability density estimates for the set of hyperparameters. $l(x)$ constitutes probabilities density estimates for hyperparameter values where the observed loss is less than y^*, and $g(x)$ constitutes the probability density estimates for hyperparameters values where the observed loss is greater than or equal to y^*. As such, using the definition provided by Bergstra et al. (2011) on p.4:

$$p(x \mid y) = \begin{cases} l(x) & \text{if } y < y^* \\ g(x) & \text{if } y \geq y^* \end{cases} \tag{13}$$

To then choose an optimal set of hyperparameters for the next trial, TPE aims to sample hyperparameter values which maximise Expected Improvement (EI). EI therefore constitutes the acquisition function of the TPE algorithm. EI is given as the expectation of the improvement given the surrogate model's distribution, which we can define using Bayes' rule:

$$EI(x) = \int_{-\infty}^{y^*} (y^* - y) p(y \mid x)\, dy = \int_{-\infty}^{y^*} (y^* - y) \frac{p(x \mid y) p(y)}{p(x)}\, dy \tag{14}$$

Rather than computing this expectation directly, Bergstra, Bardenet, Bengio, & Kégl (2011) instead point out since any hyperparameter combination must come from either $l(x)$ or $g(x)$, we can express $p(x)$ as:

$$p(x) = \gamma l(x) + (1 - \gamma) g(x) \tag{15}$$

We can see from Equation 14 that $EI(x) \propto \frac{1}{p(x)}$. Therefore, by substitution and then rearrangement:

$$EI(x) \propto \frac{1}{\gamma l(x) + (1 - \gamma) g(x)} = \left(\gamma + (1 - \gamma) \frac{g(x)}{l(x)} \right)^{-1} \tag{16}$$

We can therefore see that $EI(x)$ is maximised when $\frac{g(x)}{l(x)}$ is smallest, or conversely when $\frac{l(x)}{g(x)}$ is maximised. This latter ratio is used by the acquisition function. The function samples m sets of hyperparameter values from $l(x)$ and computes $\frac{l(x)}{g(x)}$. The sample with the largest value for $\frac{l(x)}{g(x)}$ is then chosen as the next candidate hyperparameter value set. This then updates the surrogate model and the process continues for n trials. In order to provide densities $l(x)$ and $g(x)$ to the acquisition function, an initial startup period of trials is required. By default, Optuna's TPE sampler uses 10 startup trials before performing the TPE algorithm for some n number of trials.

To actually estimate $l(x)$ and $g(x)$, TPE uses kernel density estimation (KDE - a.k.a. the Parzen-Rossenblatt window method) (Parzen, 1962; Rosenblatt, 1956). KDE is a non-parametric method of estimating a probability density function given some set of n observations $\{x_1, ..., x_n\}$. In the case of TPE, these observations are the candidate hyperparameters from previous trials.

4.5.2. *Hyperband Pruning*

Each set of candidate hyperparameter values is trained across some number of epochs as specified during training. However, it is possible to evaluate trials more efficiently than to simply let each of then run their full course of epochs. In the context of this project, the number of epochs per trial is the *resource budget*. Hyperband pruning (Li et al., 2018) is a means to efficiently allocate this resource budget across trials as to avoid wasting training epochs on trials which are clearly performing poorly.

Firstly, hyperband tuning involves three key parameters: R_{min}, R_{max} and η. R_{min} is the minimum resource to be used per trial. As mentioned, for this project this is the minimum number of epochs each trial will train with. R_{max} is the resource budget we are choosing to employ (i.e. the maximum number of epoch per trial). η is known as the reduction factor.

Hyperband tuning uses the the successive halving algorithm to assess the performance of a trial during its progress. Whilst the standard successive halving algorithm involves running trials in parallel, the implementation in Optuna involves running trial in succession and thus an alternative method is employed. For each trial, we evaluate performance after the first rung is completed (i.e. R_{min}) and compare it to the performance of previous trials at that same rung. If performance is competitive with previous trials, we continue to the next rung, which is the current rung multiplied by the reduction factor η. We continue until either performance fails to compete, in which case the trial is terminated (i.e. *pruned*), or we reach the resource budget R_{max}.

What separates hyperband tuning from standard successive halving is that different sets of rungs are employed for some trials (known as brackets). Some brackets may involve starting with a smaller amount of resource, corresponding to more exploratory behaviour, whilst others may adopt a greater amount of resource initially, corresponding to a more exploitative approach. Trials are only compared to previous trials of the same bracket. This approach allows hyperband tuning to benefit from a balance of both exploratory or exploitative approaches, whereas standard successive halving must commit to one or the other. The number of brackets s_{max} used during tuning is determined as follows:

$$s_{\text{max}} = \left\lfloor \log_{\eta}\left(\frac{R_{\text{max}}}{R_{\text{min}}}\right) \right\rfloor \tag{17}$$

4.6. Classifier Training

Decroos, Bransen, Van Haaren, & Davis (2019) use 4 different supervised classification techniques to estimate scoring and conceding probabilities, enabling VAEP calculations. This project will replicate the implementation of these same 4 techniques as closely as possible, albeit they do not provide comprehensive details on the Python version used, librarie versions, or all model hyperparameters.

Given that VAEP requires two concurrent classifiers to be trained for scoring and conceding probability estimates, respectively, and the introduction of manually engineered spatial features and spatial embeddings will result in a total of 32 individual classifier models being trained, enabling the construction of: baseline VAEP; baseline + manual features; baseline + embeddings; and baseline + manual features + embeddings models. This ablation design allows for the impact of adding spatial context on model performance and behaviour to be investigated.

4.6.1. *Logistic Regression*

Logistic regression (Cox, 1958) is a classification technique which estimates the probability that a given data instance belongs to some reference class (i.e. an action results in a goal scored/ conceded vs no goal scored/conceded, within k events). Logistic regression learns a vector of weights $\mathbf{w}$ and bias b which minimises cross-entropy loss and produces a probability estimate for the data instance beloning to the reference class:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}} \tag{18}$$

Decroos, Bransen, Van Haaren, & Davis (2019) report the use of L2 Regularisation and a Limited-memory BFGS, which is replicated in this project.

L2 regularisation introduces an additional penalty term to the loss function (in this case cross-entropy loss, see Section 4.6.4). This penality term is some multiple $\lambda \in \mathbb{R}^+$ of the sum of squared weights:

$$\text{L2} = \lambda \sum_{i=1}^{N} w_i^2 \tag{19}$$

`scikit-learn` uses a hyperparameter $C = \frac{1}{\lambda}$ to control regularisation strength, which is set to 1 by default. Since the authors do not specify otherwise, this default will be maintained for training.

Limited-memory BFGS (Liu & Nocedal, 1989) is a computationally-efficient optimisation method, used in logistic regression to iteratively minimise the loss function during training. Again, this hyperparameter is the default for `scikit-learn`.

4.6.2. *Random Forest*

Classification tasks may also be performed by decision trees. We can treat a classification task as a form of recursive partitioning of the dataset until we obtain maximally *pure* partitions primarily consisting of data instances from a single classification class. Choosing a feature and subsequent splitting threshold which minimises impurity (see Section 4.6.5) until a maximum tree depth or minimum leaf node size is reached builds a decision tree which can be used to make classifications by applying a sequence of logical filters.

A random forest (Breiman, 2001) is an ensemble method which aggregates results from multiple decision trees, each trained on a random subset of the training set. In a classification task, this aggregation is employed by majority voting. For each data instance, the class predictions made by each tree in the forest are counted up and the class with the most "votes" is outputted as the predicted class for that instance. By having class predictions be made via majority voting from multiple decision trees, a random forest benefits from being less prone to overfitting and more robust to outliers in the dataset.

Decroos, Bransen, Van Haaren, & Davis (2019) indicate that they used 100 trees and 40 parallel threads to train their random forest models. Thus this project also used 100 trees. However, the number of parallel threads simply determines the number of concurrent processes used, which does not impact model behaviour other than via its runtime. Since my hardware is different to the authors', this hyperparameter will be chosen during training.

4.6.3. *Gradient Boosting*

Gradient boosting (Friedman, 2001) is another ensemble method involving decision trees. Gradient boosting methods construct a "strong" predictor by sequentially adding "weak" learners, where each new learner is trained to correct the errors of the previous ensemble.

$$\hat{y}_i = \sum_{t=1}^{T} \eta f_t(x_i), \quad f_t \in \mathcal{F} \tag{20}$$

Each function f_t is learned sequentially to minimise a loss function (like cross-entropy loss). At each iteration, the new weak learner is fit to the negative gradient of the loss with respect to the current model predictions. The learning rate η is used to determine how much the next tree contributes to model estimates.

4.6.3.1. *XGBoost*

XGBoost (Chen & Guestrin, 2016) builds upon standard gradient boosting by incorporating regularisation and introducing improved computational efficiency, even being capable of handling sparse data (i.e. data instances which are empty in some subset of features). Decroos, Bransen, Van Haaren, & Davis (2019) use 100 trees which are allowed to reach a maximum depth of 3 nodes. The learning rate η is equal to 0.1. Again, 40 parallel threads are used. Each of these hyperparameters (except the parallel threads) will be held constant in this project.

4.6.3.2. *CatBoost*

CatBoost (Prokhorenkova et al., 2019) is a gradient boosting method which is optimised for processing categorical predictors without requiring manual one-hot encoding. Decroos, Bransen, Van Haaren, & Davis (2019) retain the default hyperparameters for CatBoost (CatBoost Developers, n.d.), and again use 40 parallel threads.

4.6.4. *Loss Function: Cross-Entropy Loss*

Logistic regression, XGBoost and CatBoost each minimise cross-entropy loss in their classification predictions, according to their default implementations in the aforementioned Python libraries. Cross-entropy measures how "surprised" the model is by observations drawn from the true data-generating distribution $p(x)$, when evaluated under the predicted distribution $q(x)$. If the model tends to assign low probability to data instances which are probable under $p(x)$, then cross-entropy will be high. A low cross-entropy is found when $q(x)$ approximates $p(x)$ well.

$$H(p, q) = -\sum_{x \in X} p(x) \log q(x) \tag{21}$$

4.6.5. *Split Criterion: Gini impurity*

Whilst the split criterion used for the random forest models is not specified, the authors do indicate that they "used the implementation from the `scikit-learn` Python package" (Decroos, Bransen, Van Haaren, & Davis, 2019, p. 1860). It is thus inferred that they use the default split criterion of Gini impurity (Pedregosa et al., 2011; scikit-learn Developers, n.d.).

Gini impurity indicates the probability of assigning an incorrect label to a randomly chosen data instance by simply using the relative frequence of each label's occurrence in the data.

$$G_m = 1 - \sum_{k=1}^{K} \left(\frac{n_{m_k}}{n_m} \right)^2 \tag{22}$$

Decision trees within random forest classifiers, at each splitting decision, choose the feature and value threshold s which minimises the sum of the weighted Gini impurities of each partition m:

$$G_{m_{<s}/m_{>s}} = \frac{n_{m_{<s}}}{n_m} G_{m_{<s}} + \frac{n_{m_{>s}}}{n_m} G_{m_{>s}} \tag{23}$$

This results in the *purest* possible split and thus corresponds to the most discriminative splitting decision available. This process then iterates on each node in the decision tree until leaf nodes are reached which each only contain a single class (in this case, 0 or 1). Data instances can then be classified according to this derived decision tree.

4.7. Model Evaluation Metrics

4.7.1. *Area Under the Receiver Operating Characteristic Curve (AUC-ROC)*

A receiver operating characteristic curve (ROC) is produced by plotting the true-positive rate (TPR) against the false positive rate (FPR) for each probability threshold a model could use to classify a data instance into the reference class (i.e. the action results in a goal scored/conceded within k events). The result is an estimate of the probability that, given a random pair of instances where one is labelled 1 and the other labelled 0, the instance labelled 1 is assigned by the classifier a higher probability of belonging to the 1 class than the instance labelled 0. A higher AUC score therefore indicates strong performance by the classifier.

$$\text{AUC} = \frac{1}{mn} \sum_{j=1}^{n} \sum_{i=1}^{m} \phi(x_i, y_i) \quad \text{where } \phi(x_i, y_j) = \begin{cases} 1 & x_i > y_j \\ 0.5 x_i = y_j \\ 0 & x_i < y_j \end{cases} \tag{24}$$

4.7.2. *Brier Score*

Decroos, Bransen, Van Haaren, & Davis (2019) use the Brier score (Brier, 1950) to measure the performance of the VAEP classifier models by comparing the model's predicted probability of a data instance belonging to a class with its true class membership.

$$\text{BS} = \frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} (p_{ik} - y_{ik})^2 \tag{25}$$

This is effectively equivalent to the mean squared error, where predictions equate to probabilities of belonging to the kth class and the true values are a one-hot encoding of that data instance's membership of that k^{th} class. For binary classification this simplifies to:

$$\text{BS}_{\text{binary}} = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2 \tag{26}$$

A lower Brier score indicates that the model's predicted probabilities are, on average, closer to the true outcomes.

4.7.3. *Cross-Entropy Loss*

Cross-entropy loss was introduced as a loss function in Section 4.6.4, but it may also be used as a classification model evaluation metric. This project intends to use cross-entropy loss to evaluate each of the two classifier models which constitute each VAEP model.

4.7.4. *Expected Calibration Error (ECE)*

Expected calibration error (ECE) (Naeini et al., 2015) is a performance metric used to evaluate the ability of a classification model to make accurate predictions. ECE creates m bins between 0 and 1 and assigns predictions to bins based on their probability estimates. For each bin, we take the accuracy of predictions in that bin o_i and subtract from it the average of the probability estimates in that bin e_i. This is weighted by the proportion of predictions in that bin. Each bin's score is the summed to provide the ECE of the model's predictions. A lower ECE is more desirable, indicating that model predictions are close to true frequencies.

$$\text{ECE} = \sum_{i=1}^{K} P(i) \cdot |o_i - e_i|$$

4.8. Model Interpretability

4.8.1. *Permutation Importance*

Permutation importance is a method for estimating feature importance by randomly permuting one or more features and evaluating the change in model performance metrics. Features which, when permuted, result in a larger drop in model performance can be inferred to be more important in model estimations. Care should be taken with multicolinearity, as permutations will not entirely disconnect their relationship with others model features since those relationships can continue to exist via their colinear counterparts. Permutation importance will thus be used in this project to highlight which features or set of features are most determinative of the VAEP values assigned to actions.

4.8.2. *Partial Dependence Plots (PDPs)*

Partial dependence plots (PDPs) (Friedman, 2001) are used to visualise the relationship between some set of features (usually just 1 or 2) and the predictions made by a machine learning model, holding other features constant. In this case, we can use PDP plots to explore our interpretable feature set, that is the baseline and manually-engineered features, as opposed to the spatial embeddings whose dimensions are not directly interpretable.

4.8.3. *k-Means Clustering*

We can also explore the spatial embeddings via k-means clustering, similarly to how DeepCluster applies k-means to image embeddings generated by a convolutional neural network (CNN) (Caron et al., 2019). k-means is an unsupervised clustering algorithm which finds optimal locations for some k centroids within the d-dimensional feature space. The algorithm initialises k centroids within the feature space, assigns data points to their nearest centroid and then recalculates the location of each centroid to the position which minimises the distance to its assigned data points. The data points are then reassigned and the algorithm iterates until convergence. A number of distance measures may be used, with euclidean distance being most common.

$$\text{dist}\ (x_i, x_n) = \sqrt{\sum_{d=1}^{D} \left(x_{i_d} - x_{n_d}\right)^2} \tag{27}$$

The benefit of using k-means clustering with the present project's spatial embeddings is that underlying relationships represented by these embeddings can be explored by comparing the properties of clusters. For instance, player position heatmaps for each cluster could be plotted to potentially reveal notable team shapes, or if certain clusters tend to involve certain action types over others.

4.8.3.1. *Choosing* k

By fitting a model to the data with varying values of k, we can use a number of metrics to evaluate how appropriate the chosen value of k is for clustering the given data points.

4.8.3.1.1. *Inertia*

The hyperparameter k can be chosen by considering the marginal reduction in *inertia* as k increases. A suitable k can be chosen at the point where this marginal reduction becomes negligible (this is known as the *elbow method*):

$$\text{Inertia} = \sum_{j=1}^{K} \sum_{x_i \in C_j} \parallel x_i - \mu_j \parallel^2 \tag{28}$$

Whilst commonly used in deriving k and a somewhat informative heuristic, the elbow method is inherently statistically arbitrary as a means for determining the appropriate number of clusters (Schubert, 2023).

4.8.3.1.2. *Average Silhouette Score*

An additional metric is the average *silhouette score* (Rousseeuw, 1987) for each value of k. For each data instance i, we can determine its average distance to others instances in the same cluster (known as *cohesion* $a(i)$) and its average distance to instances in the next nearest cluster (known as *separation* $b(i)$). Taking the differences between the separation and cohesion and dividing by the maximum of the two, we get the data instance's silhouette score.

If separation is lower than cohesion for a data instance, the silhouette score will be negative and thus suggests the data instance is closer to some other cluster(s) relative to its distance to its assigned cluster. Likewise, greater cohesion over separation suggests the data instance is appropriately assigned to its cluster, in terms of the distance metric used.

An average can then be taken across all data points and this average score can be compared across different values for k, where 1 indicates good overall clustering and -1 poor:

$$\text{SC} = \frac{1}{N} \sum_{i=1}^{N} \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{29}$$

4.8.3.1.3. *Calinski-Harabasz Index (CHI)*

We may also appeal to the Calinski-Harabasz index (CHI) (Caliński & Harabasz, 1974) in assessing a suitable k. CHI is just a ratio of the between-cluster sum of squares (BCSS) and the within-cluster sum of squares (WCSS, i.e. the inertia), where the former is the sum of squared

distances between cluster centroids and the latter is the sum of squared distances of each data point to its respective cluster. Each of these sum of squares measures are normalised by their respective degrees of freedom.

$$\text{CHI} = \frac{\frac{1}{k-1}\sum_{i=1}^{k} n_i \parallel c_i - c \parallel^2}{\frac{1}{n-k}\sum_{i=1}^{k}\sum_{x \in C_i} \parallel x - c_i \parallel^2} \tag{30}$$

4.8.3.1.4. *Davies-Bouldin Index (DBI)*

Lastly, the Davies-Bouldin index (DBI) provides a simultaneous evaluation of how separated each cluster is and how compact data instances are to their assigned cluster. For each pair of clusters i, j, we can consider how their individual compactness (S_i and S_j) and their relative distances compare by taking the ratio of the sum of their compactness to the distance of their centroids.

$$R_{ij} = \frac{S_i + S_j}{M_{ij}} \tag{31}$$

S_i and M_{ij} can, in principle, be any measure of internal compactness or cluster separation, respectively. `scikit-learn` implements the average Euclidean distance between data instances and their cluster's centroid for internal compactness, and Euclidean distance between cluster centroids for cluster separation.

A lower value for R_{ij} indicates the clusters i, j are well-separated and internally compact. As such, for each cluster i, we can consider its *worst-case scenario* by finding its corresponding j^{th} cluster which maximises R_{ij}. Taking the sum of each of these maximum R_{ij} values and dividing by the number of clusters k gives a measure of the average clustering performance $\bar{R}$, where minimising the measure optimises clustering performance:

$$\bar{R} = \frac{1}{k}\sum_{i=1}^{k} \max_{i \neq j} R_{ij} \tag{32}$$

4.8.4. *Uniform Manifold Approximation and Projection (UMAP)*

In order to visualise and explore the spatial embeddings, uniform manifold approximation and projection (UMAP) (McInnes et al., 2020) can be used as a dimensionality reduction technique to represent these embeddings in a 2 or 3 dimensional space. The color of data points within a UMAP can then be determined by properties such as the VAEP value, scoring/conceding probability or k-means clusters, allowing for the observation of emergent representational structure within the spatial embeddings, should any structure arise.

For each data point, UMAP locates its k nearest neighbours (where k is a pre-specified hyperparameter). A directed graph is constructed between each data point and its neighbours, where the weight of each directed connection is determined as follows:

$$w\big(x_i, x_{i_j}\big) = \exp\left(\frac{-\max\big(0, d\big(x_i, x_{i_j}\big) - \rho_i\big)}{\sigma_i}\right) \tag{33}$$

$d\big(x_i, x_{i_j}\big)$ is the raw distance between points i and j (e.g. Euclidean distance). ρ_i is the raw distance of the closest neighbour to point i. σ_i is a scaling parameter used to control how rapidly weights decrease as a function of distance. σ_i is uniquely determined for each data point i such that it satisties the following equivalence:

$$\sum_{j=1}^{k} \exp\left(\frac{-\max\big(0, d\big(x_i, x_{i_j}\big) - \rho_i\big)}{\sigma_i}\right) = \log_2(k) \tag{34}$$

Here, the hyperparameter k is the aforementioned number of nearest neighbours each data point is pre-specified to have. This scale parameter ensures the size of each neighbourhood is comparable between regions of varying densities.

With each directed weighted graph for each data point constructed, these graphs can then be combined into a global directed weighted graph for the entire data set. To ensure symmetry between connections from points i and j (which is not ensured when calculating $w\big(x_i, x_{i_j}\big)$), weights are updated as follows:

$$w_{ij} = w\big(x_i, x_{i_j}\big) + w\big(x_j, x_{j_i}\big) - w\big(x_i, x_{i_j}\big)w\big(x_j, x_{j_i}\big) \tag{35}$$

Each data point is then assigned an intialised lower-dimensional spectral embedding (e.g. 2 dimensions). To optimise these embeddings such that the derived lower-dimensional representation as closely resembles the original n-dimensional data set, a new global directed weighted graph of this lower-dimensional embeddings space is constructed and made to resemble the global directed weighted graph from the n-dimensional data set. The optimisation adjusts the locations of points in the lower-dimensional space such that they are positioned closer to points with which they share a stronger weighting in the original graph and further away from points with which they share a weaker weighting in the original graph. This optimisation is achieved via stochastic gradient descent, which minimises the following cross-entropy loss function:

$$\mathcal{L} = \sum_{(i,j)} w_{ij} \log\left(\frac{w_{ij}}{\hat{w}_{ij}}\right) + (1 - w_{ij}) \log\left(\frac{1 - w_{ij}}{1 - \hat{w}_{ij}}\right) \tag{36}$$

5. Results

5.1. Set Transformer Training

The set transformer used to generate spatial embeddings was trained on 652,041 atomic-spadl actions, with a validation set of 95,933 used for hyperparameter tuning. A final test set of 191,180 instances was used to test the best performing model following hyperparameter tuning. This best performing model was trained for 20 epochs, whereas models with candidate hyperparameter during tuning were trained for 10 epochs, unless pruned by the hyperband pruner.

A total of 30 trials were used for hyperparameter tuning. Table 2 shows the set of candidate values which the TPE sampler drew from. As indicated by the `Data Type` column, most hyperparameters were passed as categorical sets, whilst those which were continuous allowed the TPE sampler to sample from any value within the specified range, inclusive. The learning rate and weight decay were additionally specified to be sampled log-uniformly.

Hyperparameter	Candidate Values	Data Type
Batch Size	[256, 512, 1024]	Categorical
Dropout	[0.0, 0.1, 0.2, 0.3]	Categorical
Embedding Dimensions	[16, 32, 64]	Categorical
Flag Weight	(0.05, 0.5)	Continuous
Hidden Dimensions	[32, 64, 128, 256]	Categorical
Learning Rate	(0.0001, 0.005)	Continuous (Logarithmic)
Number of Heads	[2, 4, 8]	Categorical
Number of Hidden Layers	[1, 2, 3, 4]	Categorical
Weight Decay	(1e-06, 0.001)	Continuous (Logarithmic)
XY Weight	(0.5, 2.0)	Continuous

Table 2: Hyperparameter Candidate Values

As shown in Table 3, trial 22 performed best in terms of minimising the weighted Chamfer + BCE loss function. The hyperparameters derived by the TPE sampler are of interest here for interpreting the embeddings which were subsequently generated by this trained set transformer.

trial	value	batch size	output dim	flag weight	hidden dim	drop -out	lr	num heads	num layers	weight decay	xy weight
0	60.875	256	16	0.204	32	0.3	1.8e-03	8	3	6.8e-05	1.58
1	1437.236	512	32	0.191	128	0.1	1.1e-03	8	4	3.6e-05	1.42
2	18.687	512	32	0.351	256	0.0	1.4e-04	2	3	1.5e-04	0.52
3	518.686	256	64	0.429	256	0.1	1.6e-03	2	2	5.3e-06	1.02
4	nan	512	32	0.193	128	0.3	8.7e-04	2	1	3.6e-05	1.85
5	48.240	256	64	0.086	256	0.0	1.4e-04	4	2	1.3e-04	1.76
6	nan	512	16	0.247	32	0.1	2.4e-03	4	4	2.9e-04	1.86
7	52.487	1024	32	0.414	128	0.2	2.2e-04	2	4	2.6e-05	0.68
8	100.873	1024	16	0.407	128	0.0	6.6e-04	8	3	5.6e-04	1.88
9	107.329	1024	64	0.069	64	0.0	1.0e-04	4	3	2.5e-04	1.70
10	121.714	512	32	0.314	256	0.2	4.4e-03	2	1	1.4e-06	0.52
11	58.463	256	16	0.319	32	0.3	3.9e-04	8	3	7.7e-05	1.13
12	50.449	256	16	0.488	32	0.3	3.7e-04	8	3	8.8e-04	1.00
13	117.604	256	32	0.484	64	0.3	2.7e-04	2	3	8.5e-04	0.86
14	65.936	512	16	0.488	32	0.0	3.8e-04	8	3	3.5e-04	1.26
15	nan	256	16	0.344	256	0.3	1.8e-04	2	3	9.4e-06	0.80
16	16.950	512	32	0.372	256	0.0	4.6e-04	8	3	9.6e-04	0.60
17	26.969	512	32	0.369	256	0.0	1.0e-04	2	2	1.5e-04	0.53
18	18.106	512	32	0.284	256	0.0	5.4e-04	8	1	4.4e-04	0.69
19	25.044	512	32	0.264	256	0.0	4.9e-04	8	1	4.4e-04	0.73
20	29.878	512	32	0.281	256	0.0	6.8e-04	8	1	9.5e-04	0.93
21	28.831	512	32	0.367	256	0.0	2.5e-04	8	1	1.1e-04	0.64
22	14.629	512	32	0.297	256	0.0	5.8e-04	8	1	2.2e-04	0.59

trial	value	batch size	output dim	flag weight	hidden dim	drop -out	lr	num heads	num layers	weight decay	xy weight
23	32.188	512	32	0.230	256	0.0	5.7e-04	8	1	5.1e-04	0.68
24	39.550	512	32	0.149	64	0.2	9.5e-04	8	1	2.5e-04	0.83
25	15.982	512	32	0.296	256	0.0	4.6e-04	8	1	5.2e-04	0.65
26	36.474	1024	64	0.389	256	0.0	3.5e-04	8	1	2.0e-04	0.60
27	32.159	512	32	0.313	256	0.0	8.4e-04	4	1	6.2e-04	0.79
28	577.291	512	32	0.450	256	0.0	1.3e-03	8	4	6.8e-05	0.92
29	57.894	512	32	0.229	64	0.2	3.1e-04	8	2	3.4e-04	1.09

Table 3: Optuna Hyperparameter Tuning Results

An output dimension of 32 corresponds to the number of spatial embedding dimensions used during inference. The output dimension forces a compression of information into a fixed-dimensional representation, with additional width providing room for added complexity if appropriate, however this may also introduce the risk of losing generalisability if made too wide (Chen & Guo, 2023). Furthermore, given that these fixed-dimensional embedding representations serve as features in the downstream classification models, encouraging compression has the added benefit of contributing to dimensionality reduction and thus maintaining a parsimonious feature set.

Lee, Lee, Kim, Kosiorek, Choi, & Teh (2019) note that "Since the output of SAB [Set Attention Block] contains information about pairwise interactions among the elements in the input set X, we can stack multiple SABs to encode higher order interaction" (p.3). Here, an SAB is just a hidden layer of the set transformer encoder. As with other deep learning techniques then, additional hidden layers (SABs) generally correspond to the encoding of increased levels of abstract representation between set elements (players). Given that the TPE sampler appeared to find a value of 1 for the number of hidden layers to be optimal, it would potentially suggest that latent relationships between players within freeze-frames is relatively simple and thus does not require multiple layers of abstraction to produce embedding representation which minimise loss. This seems relatively intuitive, given that freeze-frames consist of 22 players at most.

That being said, one might suggest that this could be an artefact of the amount of training time afforded (10 epochs), whereby deeper representations require additional training time to update the additional layer parameters. However, Figure 3 shows that, after training 4 copies of the final model on a subset of the train/val/test datasets and varying just the number of hidden layers, a hidden layer size of 1 still achieves optimal performance, where it is appears that simply adding more hidden layers does not improve loss as might be expected under the afforementioned counterargument.

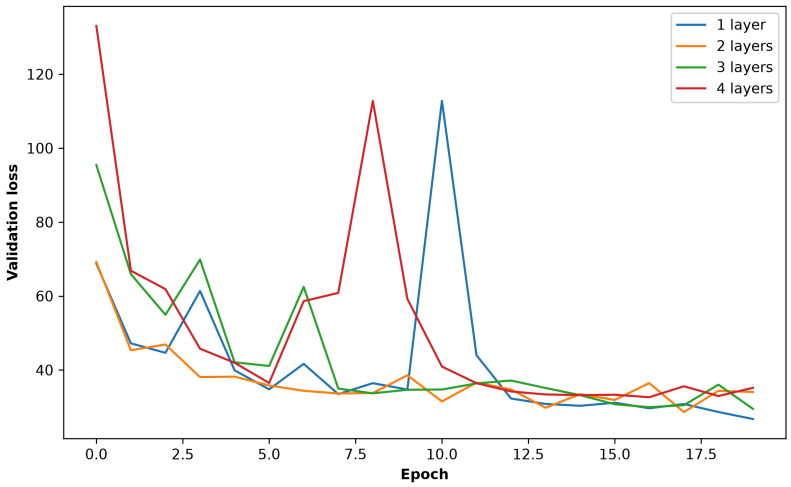


Figure 3: Comparison of Training Loss Curves for Each Value for Number of Hidden Layers [1, 2, 3, 4]

The hyperparameters from this trial were used to train the final set transformer model instance, which achieved a loss of 12.095 on the held-out test set. Figure 4 shows the training and validation loss curves for this final model. This plot shows fast convergence with good generalisability, given that both loss curves drop quickly and remain close together across the number of epochs.

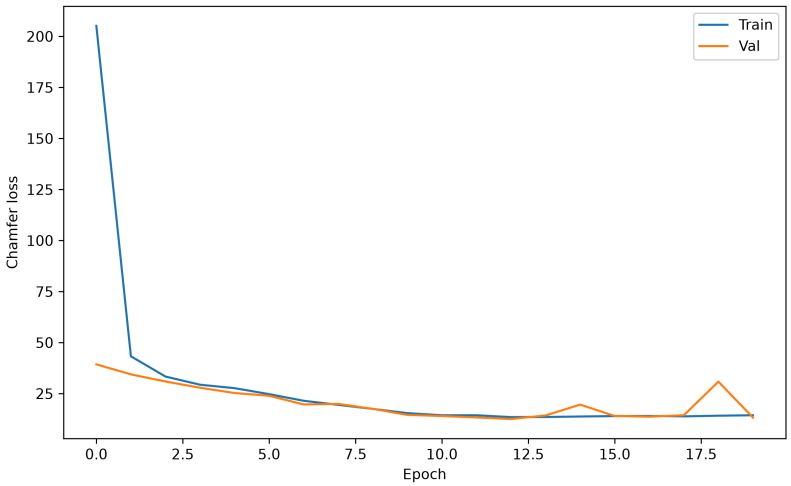


Figure 4: Set Transformer Training Loss Curve

5.2. Classifier Training

Table 4 and Table 5 show the test results from training each of the 32 classifier models.

model	feature set	target	brier score	log loss	ece	auc
CatBoost	all	concedes	0.001156	0.007513	0.000516	0.898402
CatBoost	baseline	concedes	0.001141	0.007344	0.000417	0.893438
CatBoost	embeddings	concedes	0.001187	0.007706	0.000588	0.890665
CatBoost	manual	concedes	0.001163	0.007341	0.000481	0.900236
LR	all	concedes	0.001361	0.009421	0.000190	0.786416
LR	baseline	concedes	0.001458	0.010156	0.000282	0.657340

model	feature set	target	brier score	log loss	ece	auc
LR	embeddings	concedes	0.001431	0.009595	0.000246	0.774917
LR	manual	concedes	0.001378	0.009460	0.000247	0.791045
RF	all	concedes	0.001233	0.008124	0.000177	0.902242
RF	baseline	concedes	0.001227	0.008158	0.000142	0.905104
RF	embeddings	concedes	0.001235	0.008150	0.000174	0.891494
RF	manual	concedes	0.001233	0.008162	0.000184	0.903802
XGBoost	all	concedes	0.001170	0.007099	0.000149	0.908524
XGBoost	baseline	concedes	0.001156	0.006859	0.000160	0.923141
XGBoost	embeddings	concedes	0.001172	0.007047	0.000133	0.915863
XGBoost	manual	concedes	0.001168	0.007044	0.000143	0.910457

Table 4: Concedes Classifiers Test Results

For the concede classifiers, we can see that the XGBoost classifier with the baseline feature set (i.e. *baseline*) performed best in terms of AUC (0.923) and log loss (0.007). The XGBoost classifier with the baseline + embeddings (i.e. *embeddings*) feature set showed superior calibration over other concede-based classifiers, with the *baseline* CatBoost classifier minimising Brier score.

Overall, the performance of concede classifiers showed a wider range of variation in AUC (0.657–0.923) across the 4 different classifier techniques, however if the logistic regression classifiers are excluded, this range shrinks dramatically (0.891–0.923).

Logistic regression appeared to benefit most from the introduction of spatial context features, resulting in an increase in AUC from 0.657 to 0.791 with the baseline + manually-engineered spatial context feature set (i.e. *manual*). The *embeddings* feature set also saw a large improvement in AUC, increasing to 0.775. CatBoost also saw a slight improvement in AUC with the introduction of *manual* features (0.893 to 0.900), however the opposite pattern was observed with the *embeddings* feature set.

Curiously, the random forest and XGBoost classifier techniques appeared to show a slight decrease in performance upon the introduction of spatial context features, decreasing from 0.923 and 0.910 to 0.909 and 0.891, respectively.

model	feature set	target	brier score	log loss	ece	auc
CatBoost	all	scores	0.005919	0.032045	0.001126	0.842172
CatBoost	baseline	scores	0.005967	0.033160	0.000934	0.815242
CatBoost	embeddings	scores	0.005905	0.032473	0.001238	0.827022
CatBoost	manual	scores	0.005929	0.032669	0.001161	0.827970
LR	all	scores	0.007060	0.039300	0.001501	0.755268
LR	baseline	scores	0.007231	0.042474	0.001140	0.566083
LR	embeddings	scores	0.007013	0.038910	0.001446	0.757414
LR	manual	scores	0.007051	0.039421	0.001316	0.748680
RF	all	scores	0.006568	0.035409	0.002327	0.814402
RF	baseline	scores	0.006590	0.037551	0.002019	0.743820
RF	embeddings	scores	0.006575	0.035406	0.002326	0.815046
RF	manual	scores	0.006581	0.037436	0.002155	0.758314
XGBoost	all	scores	0.005851	0.031195	0.000778	0.857998
XGBoost	baseline	scores	0.005862	0.032083	0.000728	0.828742

model	feature set	target	brier score	log loss	ece	auc
XGBoost	embeddings	scores	0.005827	0.031031	0.000817	0.856054
XGBoost	manual	scores	0.005850	0.031653	0.000736	0.842855

Table 5: Scores Classifiers Test Results

For the score classifiers, the XGBoost classifiers dominate performance. The XGBoost classifier using *all* features sets achieved the best AUC (0.858), with the *embeddings* feature set XGBoost classifier minimising the Brier score (0.006) and log loss (0.031). Furthermore, the *manual* feature set XGBoost classifier minimised calibration error most strongly (0.0007).

Overall, the performance of score classifiers showed a wider range of variation in AUC (0.755–0.856) than with the concede classifiers. Additionally, this range in score classifier AUC scores is entirely subsumed by the concede classifier AUC scores, suggesting weaker predictive performance by score classifiers relative to their concede counterparts. This is in contrast to the general pattern observed by Decroos, Bransen, Van Haaren, & Davis (2019)

5.2.1. *Feature Set Significance Tests*

The results outlined in Table 4 and Table 5 only outline point estimates for classifier performance, and thus we lack sufficient evidence to determine whether the observed variation between classifiers may just be due to random variation. In line with the research objectives, we are particularly interested in whether the observed differences in AUC scores between classifiers using different feature sets *within the same classifier type (concede vs score) and technique* vary significantly. Given that each classifier technique within each type of classifier has 4 feature sets and we want to compare them pairwise, we have $\binom{4}{2}$ unique pairwise comparisons to make (i.e. 6).

To establish whether the observed differences are actually *surprising*, under the assumption that there are no significant differences in AUC scores between classifiers, we can employ the DeLong test (DeLong et al., 1988) on each pairwise comparison of features sets within the same classifier type and technique.

The DeLong test is a non-parametric significance test used to compare the AUCs of two ROC curves which are correlated, due to being evaluated on the same set of test labels. The test estimates the covariance matrix S between the two ROC curves being compared.

$$S = \begin{bmatrix} \text{Var}\left(\widehat{AUC}_1\right) & \text{Cov}\left(\widehat{\text{AUC}}_1, \widehat{\text{AUC}}_2\right) \\ \text{Cov}\left(\widehat{\text{AUC}}_1, \widehat{\text{AUC}}_2\right) & \text{Var}\left(\widehat{AUC}_2\right) \end{bmatrix} \tag{37}$$

S can then be used to derive the variance on the difference between the two AUC scores.

$$\text{Var}\left(\widehat{\text{AUC}}_1 - \widehat{\text{AUC}}_2\right) = \text{Var}\left(\widehat{\text{AUC}}_1\right) + \text{Var}\left(\widehat{\text{AUC}}_2\right) - 2\Big(\text{Cov}\left(\widehat{\text{AUC}}_1, \widehat{\text{AUC}}_2\right)\Big) \tag{38}$$

Taking the observed difference $\Delta = \widehat{\text{AUC}}_1 - \widehat{\text{AUC}}_2$ and dividing by the standard error of this difference provides a z statistic for *Delta*

$$z = \frac{\Delta}{\sqrt{\text{Var}\ (\Delta)}}$$

Under the null hypothesis that the true population AUC scores are equal $H_0 : \text{AUC}_1 = \text{AUC}_2$, we can calculate the p-value of this observed z statistic. Provided that p is less than or equal to a pre-specified significance threshold α, we can conclude that our estimate provides sufficient evidence to reject H_0. These significance tests will employ a standard significance threshold of 0.05.

An important caveat here is that given we are employing 6 pairwise comparisons within each classifier type and technique, the family-wise error rate of these 6 comparisons will exceed α. This is because each indiviual significance test has a 5% chance of producing a false positive (concluding a significant difference when there actually isn't one). The probability of obtaining at least 1 false positive within a family of m significance tests is:

$$\text{FWER} = 1 - (1 - \alpha)^m \tag{39}$$

$m = 6$ for this project as we are making pairwise comparisons between the 4 different feature sets *within the same classifier type and technique*. This equates to a probability of obtaining a single false positive within a single family of $ = 1 - (1-0.05)^{6} $. The Holm-Bonferroni method (Holm, 1979) can be used to provide corrected p-values in light of their FWER.

This method first sorts each computed p-value for the m significance tests in ascending order and adjusts the significance threshold using the ratio $\frac{\alpha}{m+1-k}$, where k is the index (starting at 1) of each p-value in the sorted list. The first p-value is therefore testing if $p \leq \frac{\alpha}{m+1-1}$. If this is true, we move to the next p-value in the list and increment k. This continues until either the condition is not met, in which case the remaining significance tests are deemed non-significant, or all m tests have been corrected.

The implementation of signficance threshold adjustment in the `statsmodels` package involves adjusting the computed p-values themselves, rather than adjusting the significance threshold (Perktold et al., 2024; Wright, 1992, pp. 1009). Instead, we can take each p-value computed from the DeLong tests, again sorted in ascending order, and apply the following adjustment to each:

$$\tilde{p}_{(i)} = \max_{1 \leq j \leq i} \left\{ (m - j + 1) p_j \right\} \tag{40}$$

5.2.1.1. *Concedes Classifier*

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−0.64	1	4.96e-03	0.8984	0.8934
all vs embeddings	−1.09	1	7.74e-03	0.8984	0.8907
all vs manual	0.27	1	−1.83e-03	0.8984	0.9002
baseline vs embeddings	−0.31	1	2.77e-03	0.8934	0.8907
baseline vs manual	0.98	1	−6.80e-03	0.8934	0.9002
embeddings vs manual	1.03	1	−9.57e-03	0.8907	0.9002

Table 6: Concedes Model Feature Set Significance Test Comparison (CatBoost)

Table 6 shows no significant pairwise differences, indicating that the addition of spatial context yielded a non-significant change in classifier performance. We therefore lack sufficient evidence to suggest that spatial context improves the CatBoost concede classifier performance, yet we can

also say there is insufficient evidence that spatial context yields a decrease in performance for this classifier technique.

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−8.52	9e-17	1.29e-01	0.7864	0.6573
all vs embeddings	−0.93	0.4	1.15e-02	0.7864	0.7749
all vs manual	2.36	0.05	−4.63e-03	0.7864	0.7910
baseline vs embeddings	8.06	3e-15	−1.18e-01	0.6573	0.7749
baseline vs manual	8.52	9e-17	−1.34e-01	0.6573	0.7910
embeddings vs manual	1.28	0.4	−1.61e-02	0.7749	0.7910

Table 7: Concedes Model Feature Set Significance Test Comparison (Logistic Regression)

Table 7 shows 4 of the 6 pairwise comparisons yielded a statistically significant z-statistic. Results show that the addition of spatial context features, both *manual* and *embeddings*, improved logistic regression concede classifier performance substantially. Despite this, whilst a non-significant difference between *embeddings* and *manual* features sets was observed, the *manual* feature set logistic regression classifier showed a statistically significant improvement in performance compared to the logistic regression classifier using *all* feature sets, rising from 0.786 to 0.791 ($z = 2.36$, Holm-adjusted $p = .050$). This would suggest that, for the concede logistic regression classifier, the addition of spatial embeddings does not yield improved performance that is significantly greater than what is achieved by the manually-engineered feature set, and the inclusion of both feature sets results in a detriment to performance compared with what is achieved by the *manual* feature set.

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	0.50	1	−2.86e-03	0.9022	0.9051
all vs embeddings	−2.69	0.04	1.07e-02	0.9022	0.8915
all vs manual	0.33	1	−1.56e-03	0.9022	0.9038
baseline vs embeddings	−2.12	0.2	1.36e-02	0.9051	0.8915
baseline vs manual	−0.33	1	1.30e-03	0.9051	0.9038
embeddings vs manual	2.10	0.2	−1.23e-02	0.8915	0.9038

Table 8: Concedes Model Feature Set Significance Test Comparison (Random Forest)

For the random forest concede classifier, Table 8 shows just 1 statistically significant pairwise difference in classifier performance, that being *all vs embeddings* ($z = -2.69$, Holm-adjusted $p = .040$). In light of this, it would appear that the random forest concede classifier did not substantially benefit from the introduction of spatial context features.

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	2.82	0.02	−1.46e-02	0.9085	0.9231
all vs embeddings	1.67	0.3	−7.34e-03	0.9085	0.9159
all vs manual	0.44	0.7	−1.93e-03	0.9085	0.9105

comparison	z	p holm	delta auc	auc 1	auc 2
baseline vs embeddings	−2.00	0.2	7.28e-03	0.9231	0.9159
baseline vs manual	−3.40	0.004	1.27e-02	0.9231	0.9105
embeddings vs manual	−1.16	0.5	5.41e-03	0.9159	0.9105

Table 9: Concedes Model Feature Set Significance Test Comparison (XGBoost)

Table 9 shows 2 significant pairwise differences, whereby the *baseline vs manual* comparison yielded a statistically significant decrease in model performance ($z = -3.40$, Holm-adjusted $p = .004$), and the *all vs baseline* comparison shows a significant increase in classifier performance ($z = 2.82$, Holm-adjusted $p = .020$). Taken together, we find evidence of a significant reduction in XGBoost concede classifier performance due to the introduction of manually-engineered spatial features, despite a non-significant reduction due to the introduction of spatial embeddings.

5.2.1.2. *Scores Classifier*

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−5.10	2e-06	2.69e-02	0.8422	0.8152
all vs embeddings	−4.41	5e-05	1.52e-02	0.8422	0.8270
all vs manual	−3.05	0.009	1.42e-02	0.8422	0.8280
baseline vs embeddings	2.38	0.03	−1.18e-02	0.8152	0.8270
baseline vs manual	2.98	0.009	−1.27e-02	0.8152	0.8280
embeddings vs manual	0.20	0.8	−9.48e-04	0.8270	0.8280

Table 10: Scores Model Feature Set Significance Test Comparison (CatBoost)

Table 10 reveals 5 of the 6 pairwise comparisons to be statistically significant. The inclusion of *all* feature sets showed improved performance compared to the *baseline*, *embeddings* and *manual* feature sets, respectively. Both the *embeddings* ($z = 2.38$, Holm-adjusted $p = .030$) and *manual* ($z = 2.98$, Holm-adjusted $p = .009$) feature sets showed a significant improvement in performance compared to the *baseline* feature set. However, given that the *embeddings vs manual* feature sets did not yield a significant difference in CatBoost score classifier performance, it would appear that each spatial context feature set contributes to improving CatBoost score classifier performance roughly equally. Despite this, the improvement in performance observed when both feature sets are included suggests each feature set provides distinct information which the CatBoost score classifier can leverage to improve performance beyond what is achieved when either then *embeddings* or *manual* feature sets are excluded.

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−20.53	7e-93	1.89e-01	0.7553	0.5661
all vs embeddings	0.31	0.8	−2.15e-03	0.7553	0.7574
all vs manual	−7.49	2e-13	6.59e-03	0.7553	0.7487
baseline vs embeddings	19.89	2e-87	−1.91e-01	0.5661	0.7574
baseline vs manual	20.29	8e-91	−1.83e-01	0.5661	0.7487

comparison	z	p holm	delta auc	auc 1	auc 2
embeddings vs manual	−1.24	0.4	8.73e-03	0.7574	0.7487

Table 11: Scores Model Feature Set Significance Test Comparison (Logistic Regression)

Table 11 outlines DeLong test results for the logistic regression score classifier. We can observe 4 statistically significant pairwise differences. The overall finding for this classifier is that the introduction of spatial embeddings appears to improve classifier performance more substantially than that achieved by the manually-engineered feature set.

Whilst both spatial context feature sets demonstrate a substantial statistically significant improvment in classifier performance compared to the *baseline* feature set, in comparing the logistic regression score classifier using *all* features with each spatial context feature set, we see that the addition of manually-engineered features (the *all* feature set) compared to the *embeddings* feature set (i.e. the *all vs embeddings* comparison) shows no statistically significant difference in classifier performance (z 0.31, Holm-adjusted $p = .800$). However, we do observe a statistically significant decrease in classifier performance when we compare *all vs manual* ($z = -7.49$, Holm-adjusted $p < .001$). Taken together, these findings suggest that the logistic regression score classifier benefits more greatly from the introduction of spatial embeddings than manually-engineered features, in terms of classifier performance.

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−10.46	7e-25	7.06e-02	0.8144	0.7438
all vs embeddings	0.39	0.7	−6.43e-04	0.8144	0.8150
all vs manual	−7.91	1e-14	5.61e-02	0.8144	0.7583
baseline vs embeddings	10.50	5e-25	−7.12e-02	0.7438	0.8150
baseline vs manual	2.35	0.04	−1.45e-02	0.7438	0.7583
embeddings vs manual	−7.72	3e-14	5.67e-02	0.8150	0.7583

Table 12: Scores Model Feature Set Significance Test Comparison (Random Forest)

Regarding the random forest score classifier, Table 12 again shows 5 of the 6 pairwise differences to be statistically significant. Given that *all vs embeddings* yields a statistically non-significant difference in classifier performance ($z = 0.39$, Holm-adjusted $p = .700$), and the *embeddings vs manual* comparison showed a moderate statistically significant decrease in performance ($z = -7.72$, Holm-adjusted $p < .001$), we may again observe evidence that the random forest score classifier benefitted more greatly from the addition of spatial embedding features than it did from manually-engineered features.

comparison	z	p holm	delta auc	auc 1	auc 2
all vs baseline	−7.52	3e-13	2.93e-02	0.8580	0.8287
all vs embeddings	−1.27	0.2	1.94e-03	0.8580	0.8561
all vs manual	−4.58	2e-05	1.51e-02	0.8580	0.8429
baseline vs embeddings	7.36	1e-12	−2.73e-02	0.8287	0.8561
baseline vs manual	4.50	2e-05	−1.41e-02	0.8287	0.8429

comparison	z	p holm	delta auc	auc 1	auc 2
embeddings vs manual	−3.52	0.0009	1.32e-02	0.8561	0.8429

Table 13: Scores Model Feature Set Significance Test Comparison (XGBoost)

Lastly, Table 13 outlines DeLong test results for the XGBoost score classifier. Matching the same pattern observed in Table 12, we may again observe evidence that spatial embeddings provided additional information which the classifier could leverage, compared to that provided by manually-engineered features. That being said, the statistically significant difference in classifier performance between *embeddings vs manual* ($z = -3.52$, Holm-adjusted $p < .001$) is smaller (Δ AUC $= 0.0132$) than that observed for the random forest score classifier (Δ AUC $=$ 0.0567).

5.2.2. *Classifiers Used for VAEP in this Project*

Given the results outlined in Section 5.2. Section 5.2.1.1 and Section 5.2.1.2, the overall best performing classifier techniques was XGBoost, for both scoring and conceding predictions. Specifically, the *baseline* XGBoost concede classifier and the *all* features XGBoost score classifier achieved the best AUC scores. On this basis, the classifiers used to produce VAEP scores are these two classifiers. However, given that this project is, in large part, aiming to explore the performance and behaviour of spatial context features, evaluations of model interpretability will used the XGBoost score and concede classifiers trained on *all* feature sets.

5.2.3. *Valuing Actions by Estimating Probabilities (VAEP) EDA*

As seen in Decroos, Bransen, Van Haaren, & Davis (2019), it may also be useful to assess how derived VAEP values behave and align with domain-specific intuition. It is desireable for derived metrics in any domain to display strong face validity, that is, the metric should provide a well-calibrated and "plausible" (Jacobs & Wallach, 2021, pp. 5) measure of the phenomenon it is intended to capture. In measuring action value, one inherent limitation is that action-valuation metrics like VAEP lack a "ground truth" upon which to draw comparisons (Decroos et al., 2020). There is no objective measure of action value that we can observe in the world beyond what is provided by the very action-valuation metrics researchers have developed. Given that such metrics are intended to provide a quantitative measure of action value, assessing face validity is fraught with subjectivity and it is thus somewhat trivially obvious that this issue arises.

5.2.3.1. *"The Messi Test"*

One heuristic used in sports analytics is to compare derived action or player valuations with the intuitions of both technical and non-technical domain experts. For instance, Pappalardo et al. (2019) calibrate their PlayeRank metric by assessing its agreement with a panel of experienced football scouts. Regarding the period of time which publicly available events data, such as those provided by StatsBomb, covers, it is widely regarded that players such as *Lionel Messi* and *Atiana Bonmati* constitute one of the most, if not the most, valuable male and female players in world football, respectively (For example, they both lead the all-time most number of Ballon d'Or wins by a single player for their respective gender [8 and 3, respectively], Wikipedia (2026a); Wikipedia (2026b)). Dubbed "the Messi test", the heuristic suggests that an action value metric displays strong face validity if it correctly rates players who are widely regarded as being exceptionally talented as highly performing.

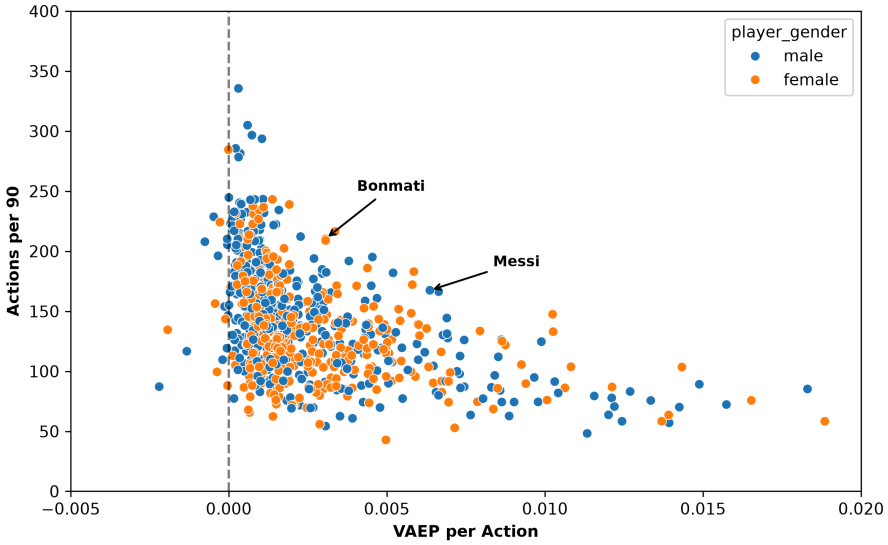


Figure 5: VAEP per Action vs Actions per 90 Minutes Played

We can see from Figure 5 that our "Messi/Bonmati test" appears to have been passed successfully. The plot includes players who have player more than 270 minutes across the competitions included in this project's dataset and compares the number of actions they perform per 90 minutes of playtime with the average VAEP value of the set of actions they took. We would expect more valuable players to exhibit both high volume (actions per 90) and high quality (VAEP per action) across their playtime and action sets. We can see that both *Messi* and *Bonmati* are positioned on the upper frontier of the scatter plot, exactly where valuable players would be expected to appear. Furthermore, the derived VAEP per action values fall within the same magnitude as that observed in Decroos, Bransen, Van Haaren, & Davis (2019), indicating some degree of agreement in valuation with previous VAEP models. The number of actions is likely generally higher than seen in Decroos, Bransen, Van Haaren, & Davis (2019) as this project uses atomic-SPADL actions, rather than standard SPADL actions.

5.2.3.2. *VAEP Player Ratings & Player Market Value*

Decroos, Bransen, Van Haaren, & Davis (2019) also explore how their derived VAEP player ratings (see Equation 3) relate to the market value of players. The authors report the top-10 rated Premier League players from the 2017/18 season along with their VAEP player ratings for that season (see Table 1d in their article).

player	minutes_played	rating
Niclas Füllkrug	276.6	1.560470
Patrik Schick	533.0	1.326983
Karim Benzema	351.6	1.232642
Wout Weghorst	448.3	1.137692
Pablo Sarabia García	272.2	1.104232
Emil Peter Forsberg	376.6	1.088416
Lionel Andrés Messi Cuccittini	732.0	1.065274
Julián Álvarez	483.9	1.055368

player	minutes_played	rating
Uğurcan Çakır	282.0	1.009587
Richarlison de Andrade	327.3	0.999384

Table 14: Top-10 VAEP Player Ratings (Male)

player	minutes_played	rating
Hinata Miyazawa	349.7	1.510352
Alexandra Popp	664.4	1.483326
Eugénie Le Sommer	366.9	1.364185
Sophie Roman Haug	281.7	1.251536
Alba María Redondo Ferrer	414.8	1.122449
Marie-Antoinette Katoto	398.0	1.099952
Sophia Olivia Smith	377.2	1.080874
Claudia Pina Medina	388.5	1.071186
Bethany Mead	708.3	1.066287
Mina Tanaka	273.0	1.060996

Table 15: Top-10 VAEP Player Ratings (Female)

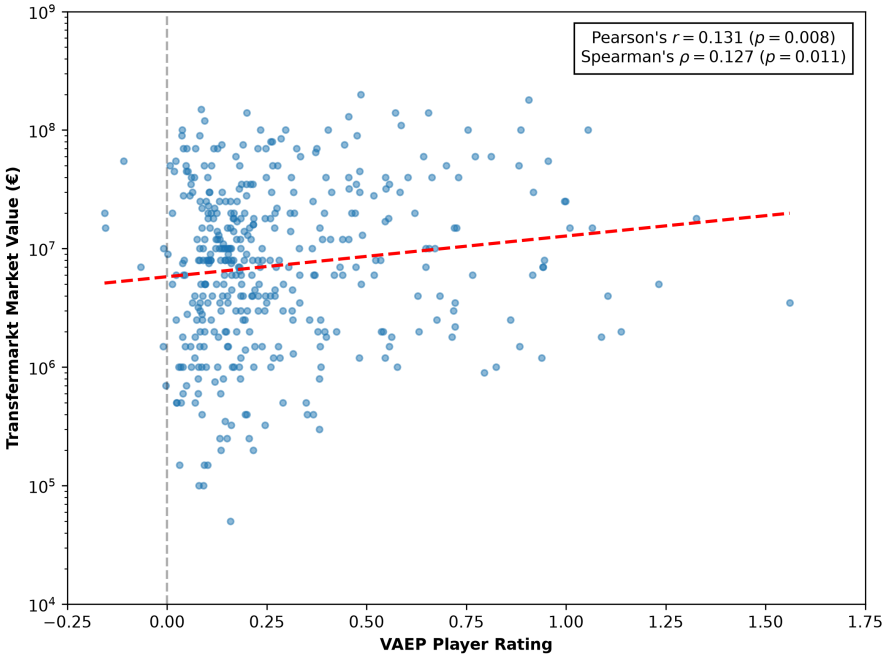


Figure 6: VAEP Player Ratings vs Transfermarkt Market Value

5.3. Model Interpretation

5.3.1. *Permutation Importance*

Beyond assessing the performance and face validity of the VAEP classifier models, we may also attempt to interpret which features are most influential in the decisions made by these classifiers. By exploring permutation importance, we can accumulate evidence to indicate which features are most impactful in enabling strong classification performance.

5.3.1.1. *Grouped Permutation Importance*

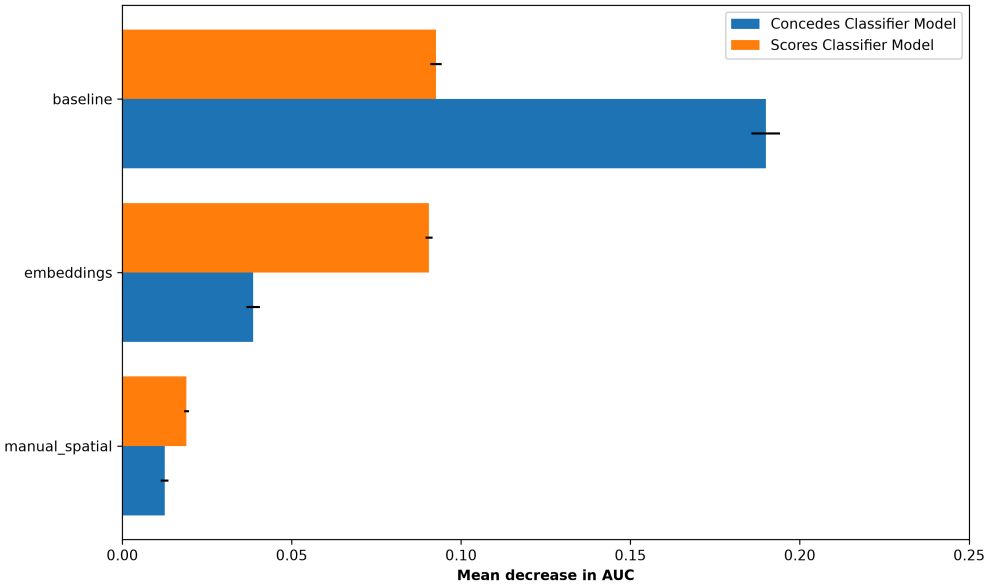


Figure 7: Grouped Permutation Importance

Figure 7 shows the average decrease in AUC (across 10 iterations) due to the random permutation of *baseline* features, spatial embeddings (*embeddings*) and manually-engineered spatial features (*manual*).

For the selected concede classifier (see Section 5.2.2), it would appear that the *baseline* feature set dominates as most important for determining the classifier's predictions, whereby the random permutation of each of these baseline features in tandem results in a reduction in AUC of around 0.18. By contrast, the *embeddings* and *manual* feature sets exhibit a much smaller reduction in AUC when randomly permuted. That being said, the random permutation of the *embeddings* and *manual* feature sets does still result in a positive-valued reduction in AUC, suggesting that these features do carry useful classification information despite Table 4 showing the addition of these feature sets to reduce classifier AUC.

Regarding the score classifier, an alternative pattern emerges. We can see that the reduction in AUC due to random grouped permutation of the *baseline* and *embeddings* feature sets is roughly identical, at around 0.09. The *manual* feature set appears much less important than these other two, given that is exhibits an average reduction of around 0.02.

Contrasting the grouped permutation results of the two classifiers, we see that spatial context features (specifically the spatial embeddings) appear much more important in determining classification decisions for the score classifier than for the concede classifier. This, of course, aligns with the observations made in Table 13. Such findings would suggest that the information represented by spatial embeddings captures spatial context relevant to informing the probability

of a given action resulting in a goal being scored and carries very little information which informs the chance of conceding from a given action.

5.3.1.2. *Baseline Feature Set Permutation Importance*

We can zoom in on each individual feature part of the *baseline* feature set to examine which features carry importance for each prediction task (score or concede). Given that each feature in the model has an additional 9 lagged variants (in order to reflect the $k = 10$ previous actions), each feature will be very highly correlated with its own lagged variants. As such, it would be more appropriate to group together each feature with its lagged variants before evaluating the permutation importance of each feature.

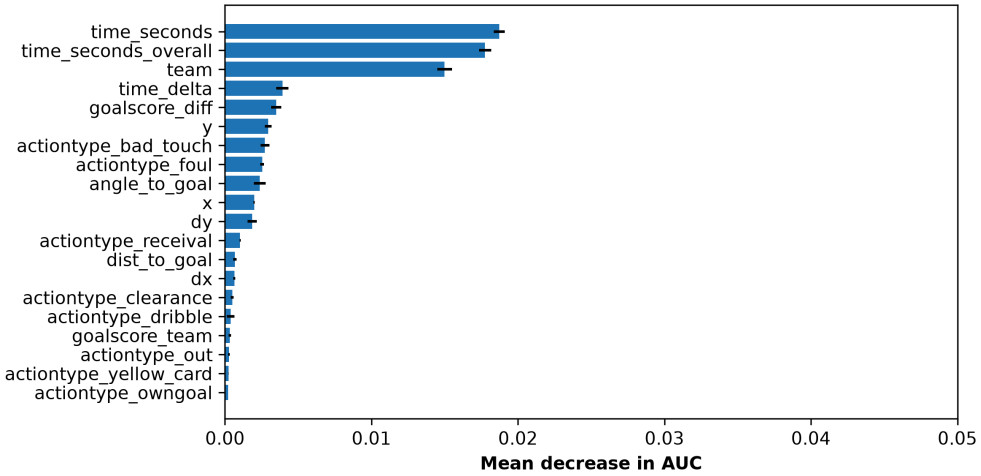


Figure 8: Concede Classifier Permutation Importance: Baseline Features

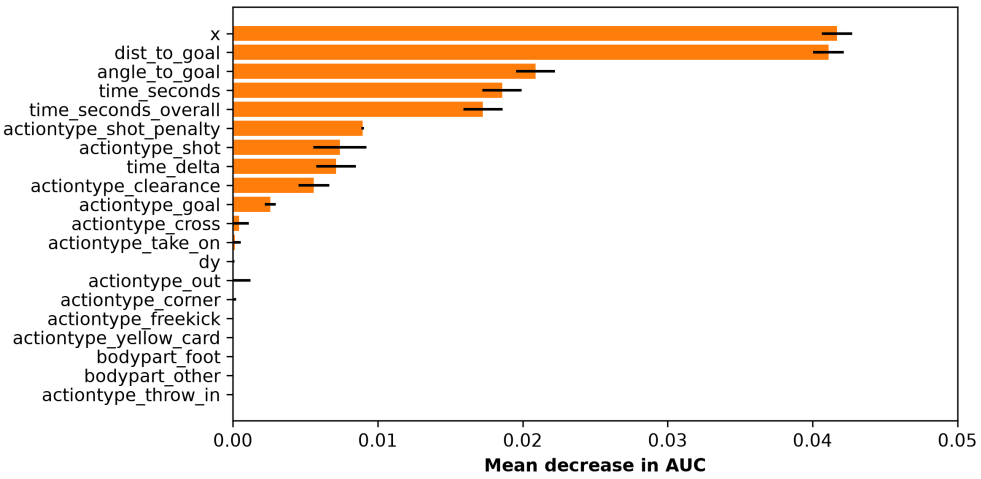


Figure 9: Score Classifier Permutation Importance: Baseline Features

5.3.1.3. *Manually-Engineered Feature Set Permutation Importance*

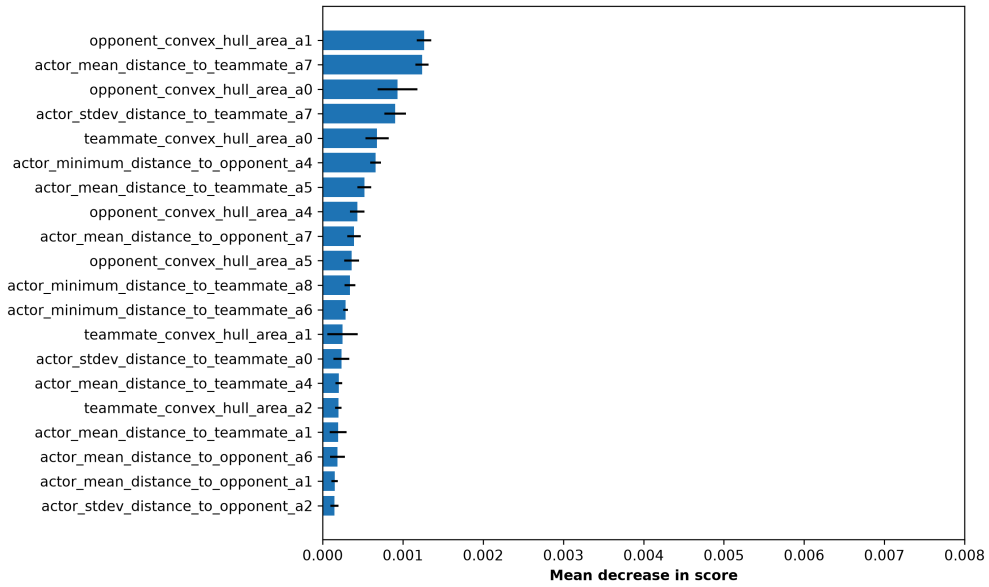


Figure 10: Concede Classifier Permutation Importance: Manually Engineered Spatial Features

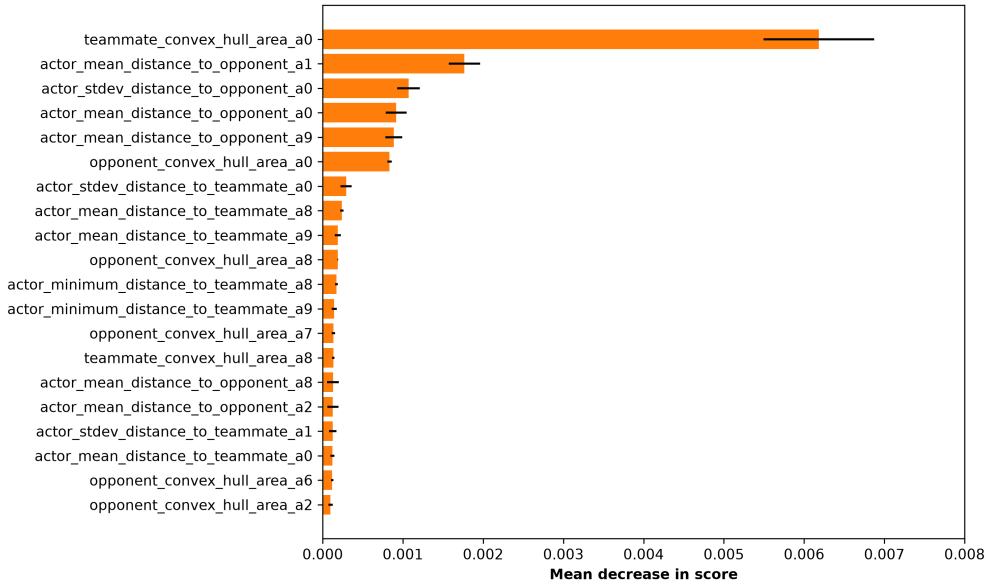


Figure 11: Score Classifier Permutation Importance: Manually Engineered Spatial Features

5.3.1.4. *Spatial Embedding Dimensions Permutation Importance*

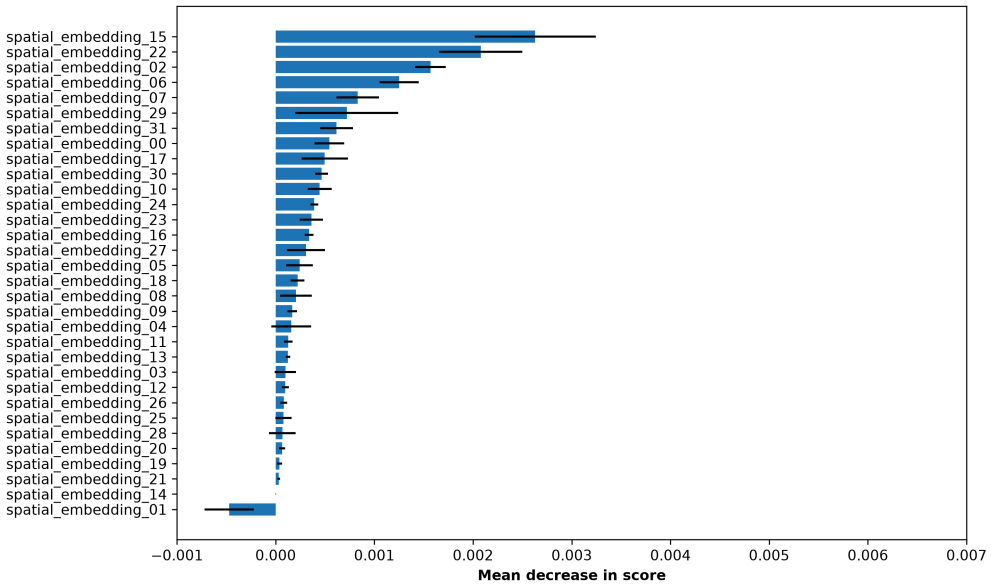


Figure 12: Concede Classifier Permutation Importance: Spatial Embedding Dimensions

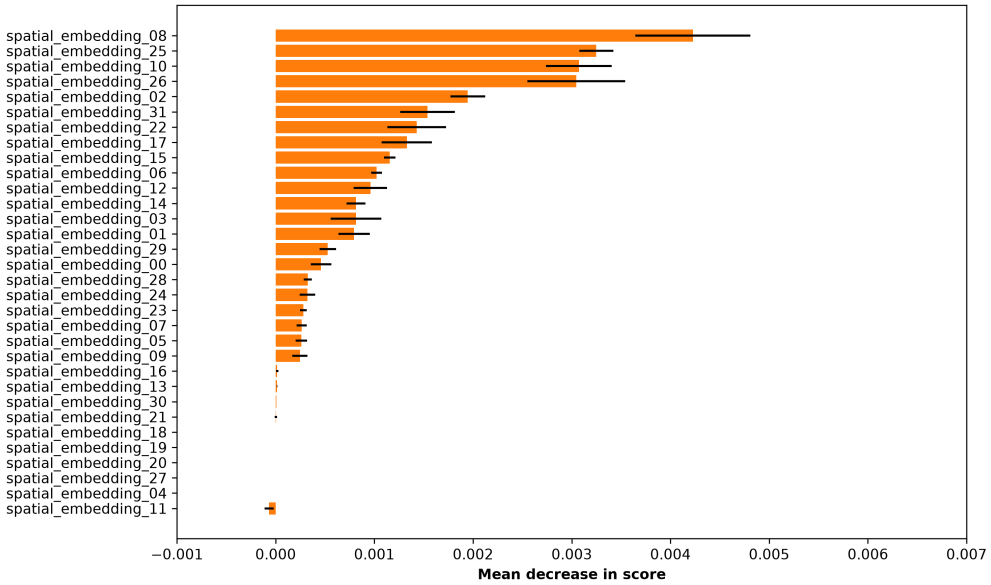


Figure 13: Score Classifier Permutation Importance: Spatial Embedding Dimensions

5.3.2. *Partial Dependence Plots*

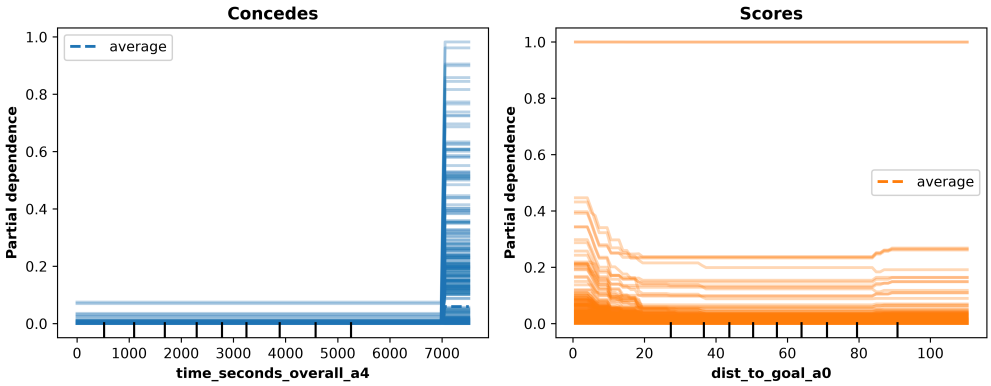


Figure 14: PDP of Most Important Baseline Features for each Model

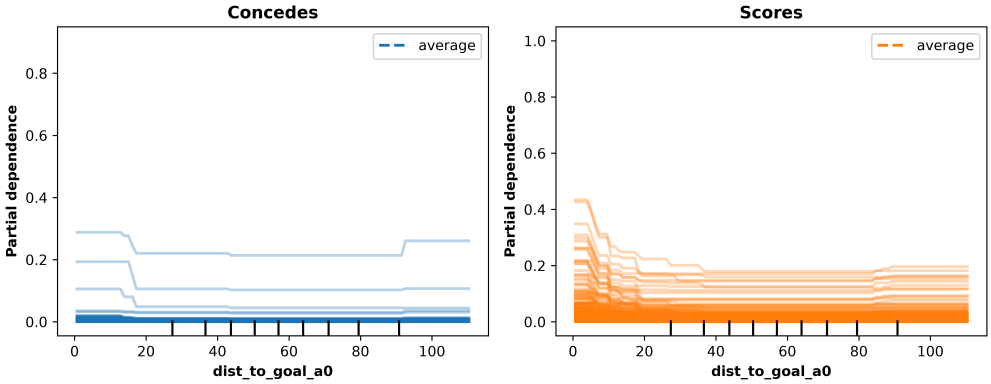


Figure 15: PDP of Distance To Goal

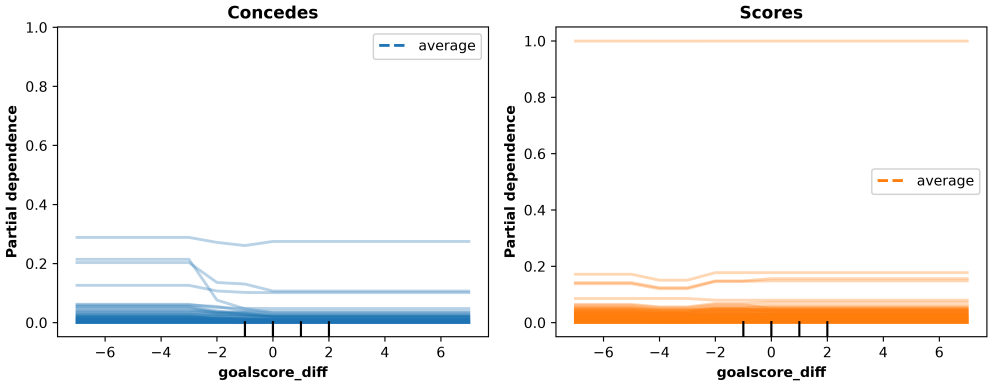


Figure 16: PDP of Goalscore Difference

5.3.3. *K-means Clustering: Spatial Embeddings*

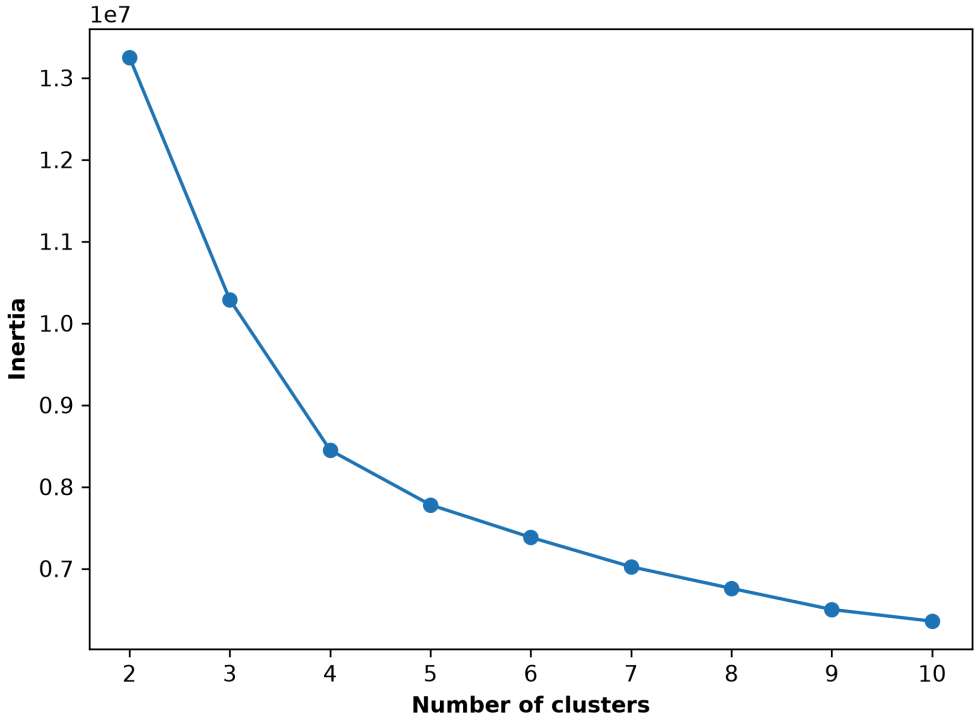


Figure 17: K-means Elbow Method Plot

k	inertia	silhouette	calinski harabasz	davies bouldin
2	1.33e+07	0.38	5.97e+05	1.08
3	1.03e+07	0.37	4.92e+05	1.03
4	8.45e+06	0.34	4.53e+05	1.25
5	7.78e+06	0.29	3.85e+05	1.58
6	7.38e+06	0.28	3.33e+05	1.70
7	7.02e+06	0.28	2.98e+05	1.76
8	6.76e+06	0.27	2.70e+05	1.85
9	6.50e+06	0.27	2.49e+05	1.88
10	6.36e+06	0.26	2.28e+05	1.92

Table 16: K-means Hyperparameter Results

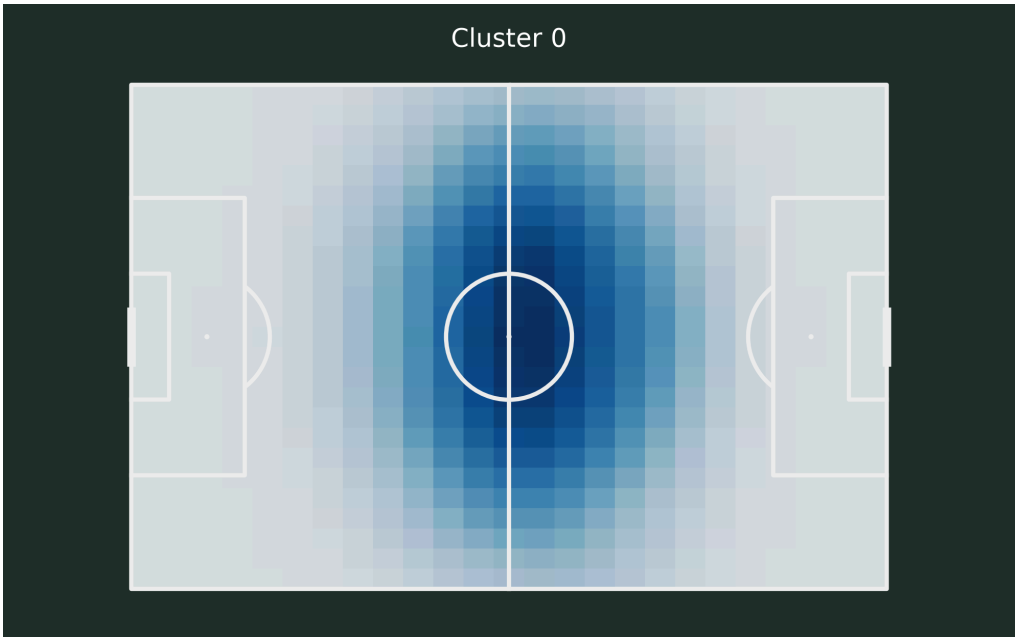


Figure 18: Pitch Heatmap of Cluster 0

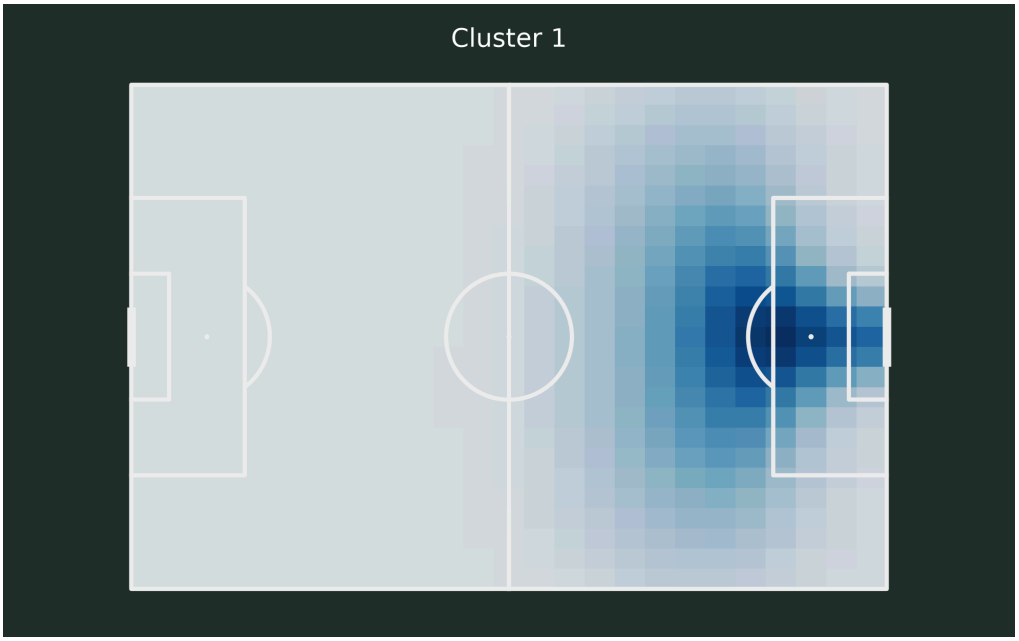


Figure 19: Pitch Heatmap of Cluster 1

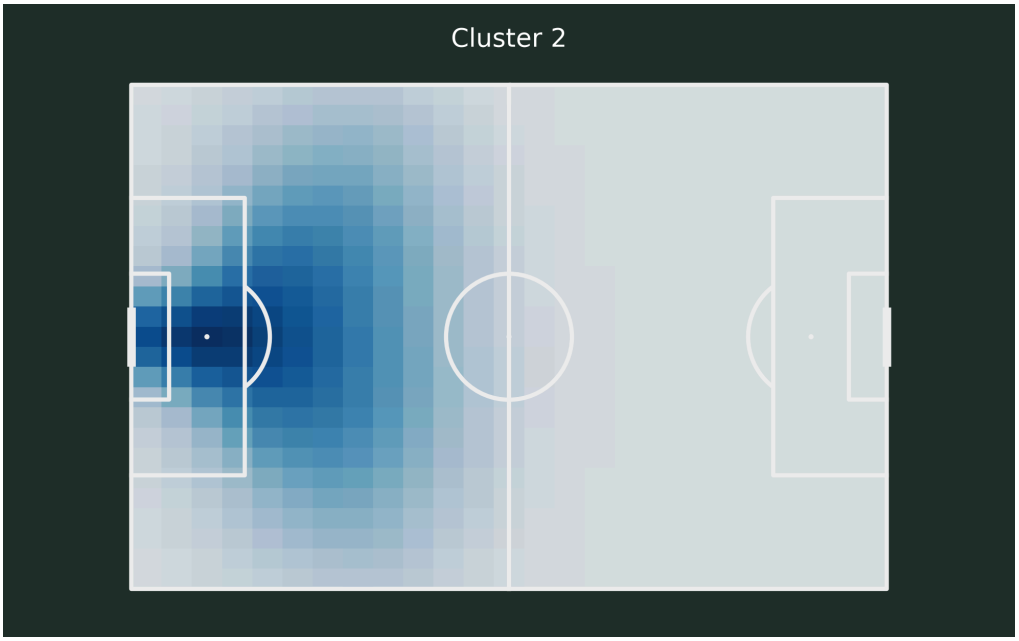


Figure 20: Pitch Heatmap of Cluster 2

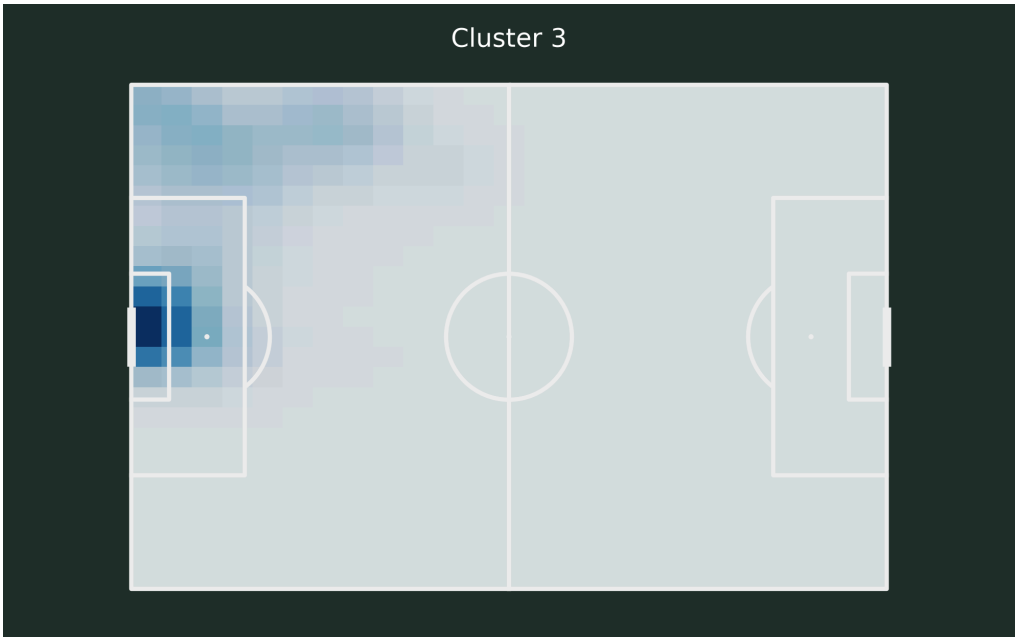


Figure 21: Pitch Heatmap of Cluster 3

5.3.4. *UMAP*

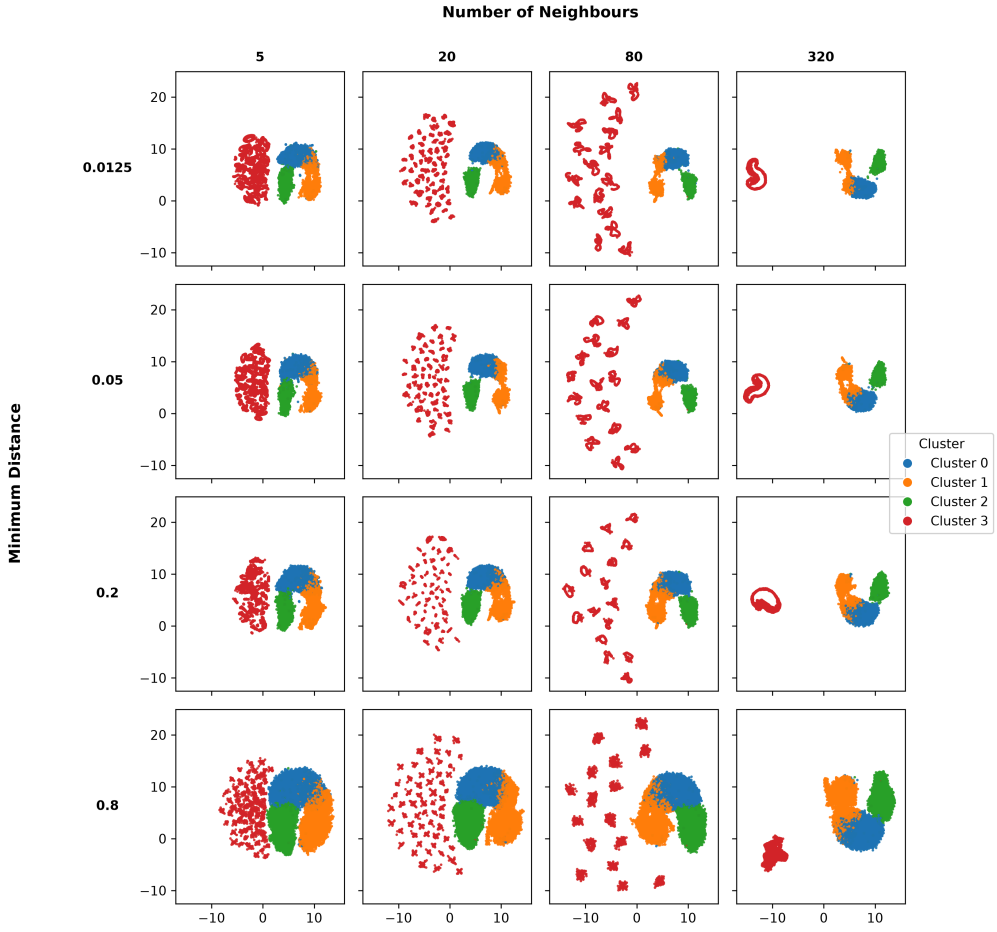


Figure 22: UMAP Coloured by K-means Cluster with Varying UMAP Hyperparameter Values

5.3.5. *Spatial Embedding Dimension and Manually-Engineered Feature Correlations*

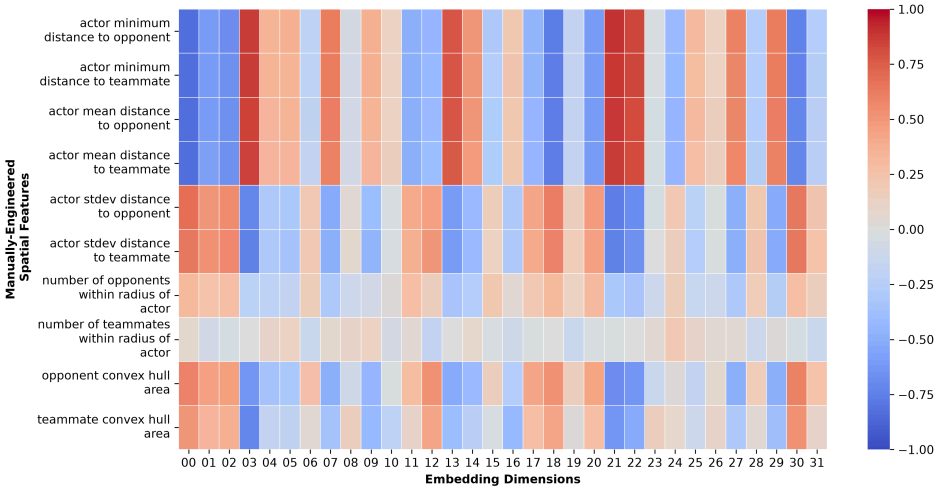


Figure 23: Manually-Engineered Spatial Features & Spatial Embeddings Correlation Heatmap

6. Discussion

7. Conclusion

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